

Ultimate Cheatsheets for ML/DL

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Contents

1	Introduction	6
1.1	Artificial Intelligence (AI)	6
1.1.1	AI Introduction	6
1.1.2	Types of AI	7
1.2	Machine Learning (ML):	8
1.2.1	ML Introduction:	8
1.2.2	Types of Machine Learning:	9
1.2.3	Supervised Learning	10
1.2.4	Unsupervised Learning	10
1.2.5	Reinforcement Learning	11
2	Algorithms	13
2.1	Regression Methods	13
2.2	Classification Methods	14
2.3	Clustering	15
2.4	Dimensionality Reduction	16
2.5	Kernel Methods	18
2.5.1	Motivation and Intuition	18
2.5.2	The Kernel Trick	19
2.5.3	Common Kernel Functions	19
2.5.4	Kernel SVM as a Primary Example	20
2.5.5	Advantages and Limitations	20
2.5.6	Kernel Methods in Modern Machine Learning	20
2.5.7	Summary	21
3	Models	22
3.1	Neural Network Architectures	22
3.2	Feedforward Networks such as Multi Layer Perceptron (MLP)	23
3.2.1	Structure and Forward Propagation	24
3.2.2	Mathematical Formulation	24
3.2.3	Training Process	25
3.2.4	Applications and Limitations	25
3.3	Convolutional Neural Networks (CNN)	25
3.3.1	Motivation and Overview	26
3.3.2	Convolutional Operations	26
3.3.3	Architectural Components	27
3.3.4	Training and Optimization	27
3.3.5	Applications and Impact	28

3.3.6	One-Dimensional Convolutional Neural Networks (1D-CNN)	28
3.4	Recurrent and Sequential Models (RNN, LSTM, GRU)	30
3.4.1	Motivation and Overview	30
3.4.2	Vanilla Recurrent Neural Networks (RNN)	30
3.4.3	Long Short-Term Memory Networks (LSTM)	31
3.4.4	Gated Recurrent Units (GRU)	32
3.4.5	Training and Practical Considerations	33
3.4.6	Applications	33
3.4.7	Summary	33
3.5	Transformer Architectures and Attention Mechanisms	34
3.5.1	Motivation and Overview	34
3.5.2	Self-Attention Mechanism	34
3.5.3	Multi-Head Attention	35
3.5.4	Positional Encoding	35
3.5.5	Transformer Encoder and Decoder	36
3.5.6	Feedforward Networks in Transformers	36
3.5.7	Training, Optimization, and Scalability	37
3.5.8	Applications and Impact	37
3.5.9	Summary	37
4	Core Components and Techniques	38
4.1	Activation Functions (ReLU, GeLU, Sigmoid, Tanh)	38
4.1.1	Theoretical Role of Activation Functions	38
4.1.2	Rectified Linear Unit (ReLU)	38
4.1.3	Gaussian Error Linear Unit (GeLU)	39
4.1.4	Sigmoid Function	40
4.1.5	Hyperbolic Tangent (Tanh)	41
4.1.6	Comparison and Practical Recommendations	42
4.1.7	Summary	43
4.2	Linear and Dense Layers (e.g., <code>nn.Linear</code>)	43
4.2.1	Definition and Mathematical Formulation	43
4.2.2	Role Within Neural Architectures	44
4.2.3	Dimensionality and Shape Considerations	44
4.2.4	Initialization and Trainability	45
4.2.5	Regularization and Stabilization Techniques	45
4.2.6	Example: A Simple Two-Layer MLP Block	45
4.2.7	Summary	45
4.3	Normalization Layers (BatchNorm, LayerNorm, GroupNorm)	46
4.3.1	Motivation for Normalization	46
4.3.2	Batch Normalization (BatchNorm)	47
4.3.3	Layer Normalization (LayerNorm)	47
4.3.4	Group Normalization (GroupNorm)	48
4.3.5	Comparison and Practical Guidelines	48
4.3.6	Summary	49
4.4	Dropout and Regularization Techniques	49
4.4.1	The Problem of Overfitting	49
4.4.2	Dropout	49
4.4.3	Weight Decay (L2 Regularization)	50

4.4.4	Early Stopping	51
4.4.5	Data Augmentation	51
4.4.6	Label Smoothing	52
4.4.7	Other Regularization Strategies	52
4.4.8	Choosing the Right Regularization Technique	52
4.4.9	Summary	53
4.5	Residual Connections and Skip Layers	53
4.5.1	Core Idea	53
4.5.2	Types of Skip Connections	54
4.5.3	Benefits	54
4.5.4	Use in Modern Architectures	54
4.5.5	Summary	54
4.6	Weight Initialization and Parameterization	55
4.6.1	The Role of Initialization	55
4.6.2	Key Initialization Principles	55
4.6.3	Xavier/Glorot Initialization	56
4.6.4	Kaiming/He Initialization	56
4.6.5	Orthogonal Initialization	56
4.6.6	Initialization for Convolutional Layers	57
4.6.7	Initialization in Transformers and Large Models	57
4.6.8	Parameterization Techniques	57
4.6.9	Summary	57
4.7	Training and Optimization	58
4.7.1	Why Model Optimization Matters	58
4.7.2	Core Components of Training	58
4.7.3	Removing Unused or Non-Connected Layers	59
4.7.4	Training Stability and Convergence	59
4.7.5	Model Compression and Efficiency Techniques	60
4.7.6	Example: Optimizing a Model for RISC-V Deployment	61
4.7.7	Summary	61
4.8	Loss Functions (Cross-Entropy, MSE, etc.)	62
4.8.1	Mean Squared Error (MSE)	62
4.8.2	Mean Absolute Error (MAE)	63
4.8.3	Huber Loss	63
4.8.4	Cross-Entropy Loss	64
4.8.5	Kullback–Leibler Divergence (KL Divergence)	65
4.8.6	Contrastive and Triplet Losses	65
4.8.7	Dice Loss and IoU Loss	65
4.8.8	Why Choosing the Right Loss Matters	65
4.8.9	Summary	66
4.9	Optimization Algorithms (SGD, Adam, RMSProp)	66
4.9.1	From Gradient Descent to Stochastic Updates	66
4.9.2	SGD (with Momentum and Nesterov)	66
4.9.3	RMSProp	67
4.9.4	Adam (and AdamW)	68
4.9.5	Choosing Between SGD, RMSProp, and Adam	69
4.9.6	Hyperparameter Heuristics and Practical Tips	69
4.9.7	Intuition via a Simple 2D Bowl	70

4.9.8	Summary	70
4.10	Learning Rate Scheduling	70
4.10.1	Theoretical Overview of Learning Rate Scheduling	71
4.10.2	Influence of the Learning Rate on Model Training	72
4.10.3	Practical Considerations	72
4.11	Early Stopping and Regularization During Training	73
4.11.1	The Dynamics of Overfitting	73
4.11.2	Implementation Strategy: Patience and Delta	74
4.11.3	Early Stopping as Implicit Regularization	74
4.11.4	Interaction with Other Techniques	75
4.11.5	Summary	75
4.12	Gradient Clipping and Stability Techniques	75
4.12.1	The Exploding Gradient Problem	75
4.12.2	Gradient Clipping Strategies	75
4.12.3	Practical Implementation and Use Cases	76
4.12.4	Summary	77
4.13	Transfer Learning and Fine-Tuning	77
4.13.1	Core Paradigms	77
4.13.2	Strategic Guidelines	78
4.13.3	Implementation Techniques	78
4.13.4	Summary	79
4.14	Implementation Patterns and Practical Considerations	79
4.14.1	Reproducibility and Seeding	79
4.14.2	Device-Agnostic Code Patterns	80
4.14.3	Numerical Stability Patterns	80
4.14.4	Gradient Accumulation	80
4.14.5	Summary	81
4.15	Model Definition in PyTorch (Modules, Forward Pass, Parameters)	81
4.15.1	The <code>nn.Module</code> Anatomy	82
4.15.2	Parameters and Buffers	82
4.15.3	Example: A Simple MLP Block	83
4.15.4	The Functional API vs. Module API	83
4.15.5	Summary	84
4.16	Training Loops and Evaluation Pipelines	84
4.16.1	The Standard Training Loop	84
4.16.2	The Evaluation Pipeline	84
4.16.3	Implementation Example	85
4.16.4	Best Practices	86
4.16.5	Summary	86
4.17	Saving, Loading, and Deployment of Models	86
4.18	Performance Profiling and Debugging	86
5	Metrics & how to read it	87
5.1	Accuracy & Loss	87
5.1.1	Accuracy	87
5.1.2	Loss	88
5.2	Precision & Recall	88
5.2.1	Precision	89

5.2.2	Recall	89
5.2.3	Confusion Matrix	90
5.2.4	Precision–Recall Trade-off and Curve Interpretation	91
5.3	F1-Score	91
6	Implementation & Code Patterns	93
7	Advanced Topics	94

Chapter 1

Introduction

This document serves as a comprehensive cheat sheet for studying Machine Learning and Deep Learning. It provides a structured summary of key algorithms, models, technologies, and optimization techniques, covering both foundational and state-of-the-art approaches.

The goal is to offer a concise yet rigorous reference for students, researchers, and practitioners seeking to review or deepen their understanding of modern artificial intelligence methodologies, including theoretical principles, implementation strategies, and code-level insights.

1.1 Artificial Intelligence (AI)

In this section we talk about Artificial Intelligence, in this chapter we talk in general about AI as we dive more deep later.

1.1.1 AI Introduction

Artificial Intelligence (AI) refers to the ability of machines or computers to mimic the way humans think and make decisions. AI enables machines to think like humans or to replicate certain functions of the human brain. In simple words, **AI is when we enable computers to think.**

Artificial Intelligence allows computers to understand, analyze data, and make decisions without human help or interaction. We use AI systems every day. A few common examples are *Siri*, *Alexa*, and *ChatGPT*.

Other categories where AI is used include:

- Virtual Assistants
- Social Media Algorithms
- Online Shopping Recommendations
- Predictive Text and Autocorrect
- Healthcare Diagnostics
- Language Translation Services
- Fraud Detection in Banking

AI enables computers to become intelligent with numbers and rules, performing calculations at incredible speed and with perfect accuracy. On the other hand, humans possess not only reasoning but also emotions, creativity, and the ability to adapt to complex and unpredictable situations.

1.1.2 Types of AI

Artificial Intelligence is divided based on two main categorization — based on capabilities and based on functionality of AI.

The following image illustrates these types of AI:

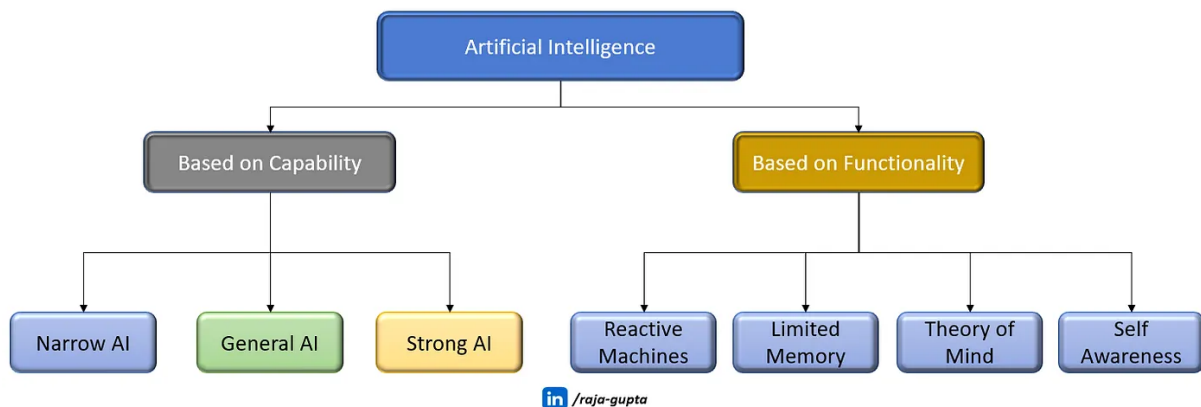


Figure 1.1: Types of AI

Now let's dive in the details of Narrow AI, General AI and Super AI.

- **Narrow AI:** Narrow AI, also known as Weak AI, refers to **artificial intelligence systems that are designed and trained for a specific task** or narrow set of tasks. Narrow AI is focused on performing a single task extremely well, but it cannot perform beyond its field or limitations.
- **General AI:** General AI, also known as Strong AI or artificial general intelligence (AGI), **can understand and learn any intellectual task that a human being can.**
- **Super AI:** Super AI represents a degree of intelligence in systems where **machines have the potential to exceed human intelligence**, outperforming humans in tasks and exhibiting cognitive abilities.

We can also divide types of AI based on functionality:

1. **Reactive Machines:** Reactive machines are AI **systems that have no memory**. These systems operate solely based on the present data, taking into account only the current situation. They can perform a narrowed range of pre-defined tasks.
2. **Limited Memory:** As the name indicates, Limited Memory AI can take informed and improved decisions by looking at its past experiences stored in a temporary memory.

This AI doesn't remember everything forever, but it uses its short-term memory to learn from the past and make better decisions for the future.

A **good example of Limited Memory AI is Self-driving cars**. The AI system in self-driving car utilizes recent past data to make real-time decisions. For instance, they employ sensors to recognize pedestrians, steep roads, traffic signals, and more, enhancing their ability to make safer driving choices. This proactive approach contributes to preventing potential accidents.

3. **Theory of Mind:** This is similar to Super AI — We should pray that we don't reach the state of AI, where **machines have their own consciousness and become self-aware**.

Self-aware AI systems will be super intelligent, and will have their own consciousness, sentiments, and self-awareness. They will be smarter than human mind.

4. **Self-Aware AI:** This is similar to Super AI — We should pray that we don't reach the state of AI, where machines have their own consciousness and become self-aware.

Self-aware AI systems will be super intelligent, and will have their own consciousness, sentiments, and self-awareness. They will be smarter than human mind.

As shown in movie "I, Robot," an AI system named VIKI becomes self-aware and starts making decisions to "protect" humanity in a controversial way.

Similar to Theory of Mind, Self-aware AI also does not exist in reality. Many experts, for example Elon Musk and Stephen Hawking have consistently warned us about the evolution of AI.

1.2 Machine Learning (ML):

In this chapter we are talking about Machine Learning, we first start explaining machine learning from a kids prospective and later we add more details.

1.2.1 ML Introduction:

Imagine we want to **enable the robot to identify several animals**.

To do so, we will show him pictures of various dogs, cats, bunnies and other animals and **label each picture with the name of the animal**. We train the robot to identify animals based on size, color, body shape, sound etc.

Once the training is completed, the robot will be able to identify these animals we trained him for.

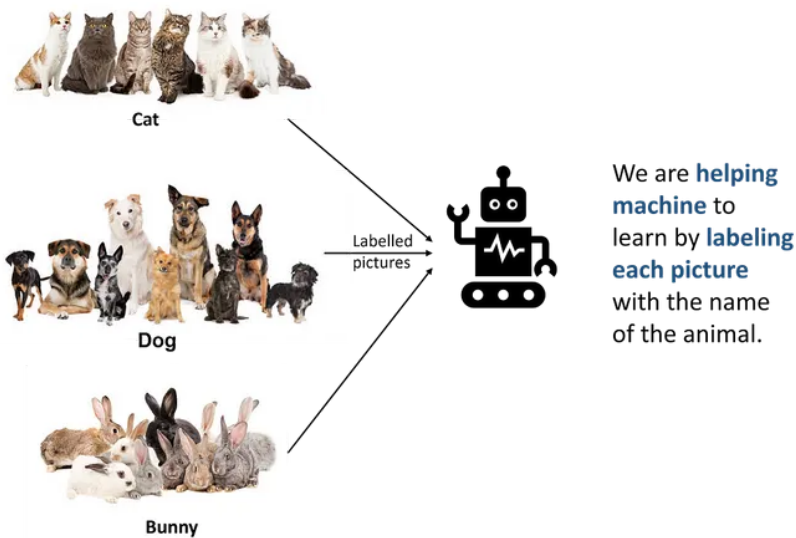
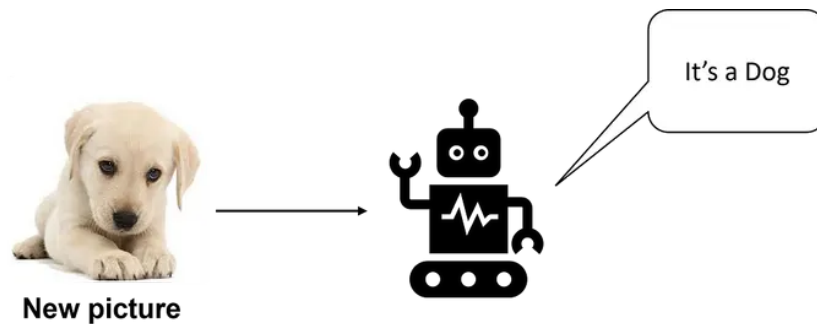


Figure 1.2: Machine Learning scheme



The machine identified the dog even though this picture is not exactly same as any of the dog's picture shown earlier.

The machine learned to identify a dog.

Figure 1.3: After training scheme

All dogs do not look alike. However, once robot has seen many pictures of dogs, **it can identify any dog even if it does not exactly look like a specific picture**. We need to show lots of pictures of dog to the robot. More pictures it sees, more efficient it will be.

In simple words Machine Learning is teaching a robot or a machine by giving a lot of examples to work with and learn.

1.2.2 Types of Machine Learning:

Machine Learning can be categorized into three main types:

1. Supervised Learning.
2. Unsupervised Learning.

3. Reinforcement Learning.

Each type serves different purposes and involves different approaches to learning from data. Let's have a close look into all these types.

1.2.3 Supervised Learning

When we trained our robot by showing pictures of animals, we labelled each picture with the name of the animal. So, **we acted as a teacher to him**. We first told him how does a dog or a cat look like and then only he was able to identify them.

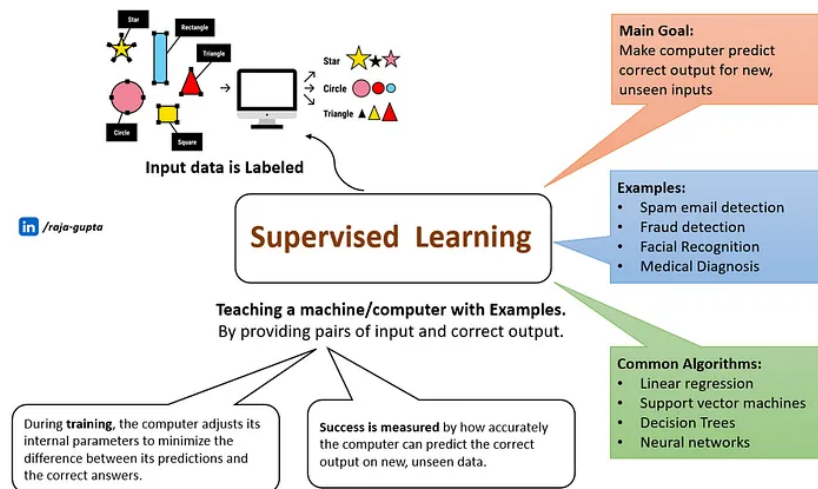


Figure 1.4: Supervised Learning summarized

Supervised Learning is widely used in this field: *Email Spamming Filtering, Image Classification, Facial Recognition, Financial Fraud Detection, Speech Recognition*.

1.2.4 Unsupervised Learning

Let's understand this from a kid's school example. When kids go to their class first day, they meet lots of classmates. At first all classmates are same to them. But with time, they themselves categorized them in different groups:

- They find some classmates very good and want to be friend with them.
- They find some rude or irritating and want to avoid them.
- They find some very good in sports and want to be in the same team as they are.
- and so on ...

When kids categorized their classmates, nobody told them how to do that. They did that without anyone's help. — This is how unsupervised learning works.

Let's take a proper machine learning example. Imagine we showed lots of pictures of dogs, cats, bunnies etc. **without any label** to our robot and told him — "*I'm not going to tell you which one is which. Go explore and figure it out*".

The robot starts to look at these animals, noticing things such as their fur, size, and how they move. It doesn't know their names yet, but it's trying to find patterns and differences on its own.

After exploring, the robot might notice that:

- Some animals have long ears (bunnies)
- Some animals have soft fur and tail (cats)
- Some animals have wagging tails (dogs)

In the end, the robot **might not know the names of the animals**, but it can say that **“These animals are similar in some ways, and those are different in other ways.”** — This is Unsupervised Learning.

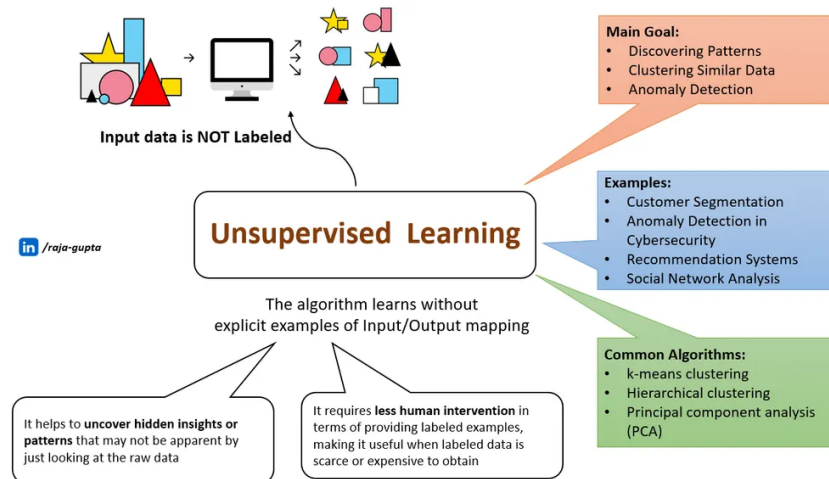


Figure 1.5: Unsupervised Learning summarized

Unsupervised Learning is widely used in this field: *Clustering Customer Segmentation, Anomaly Detection in Cybersecurity, Recommendation System.*

1.2.5 Reinforcement Learning

Imagine teaching a dog a new trick — you **reward it with a treat when it does the trick correctly** and give no treat when it doesn't. Over time, the dog **learns to perform the trick to get more treats**.

Similarly, Reinforcement Learning is:

- Training a computer to make a decision.
- By rewarding good choices and punishing bad ones.
- Just as you might train a dog with treats for learning tricks.

In reinforcement learning, there's an agent (for example a robot or computer program) that interacts with an environment. Let's take an example of teaching a computer program to play a game, for example chess.

- In this case, computer program is agent and chess game is the environment.
- The computer program can make different moves in the game, such as moving a chess piece.

- After each move, it receives feedback (reward or penalty) based on the outcome of the game.
- If it loses the game, it receives a negative reward, or a 'penalty'.
- Through trial and error, the program learns which moves lead to the best rewards, helping it figure out the best sequence of moves that leads to winning the game.

Reinforcement learning is powerful because it **allows machines to learn from their experiences** and make decisions in complex, uncertain environments — similar to how we learn from trial and error in the real world.

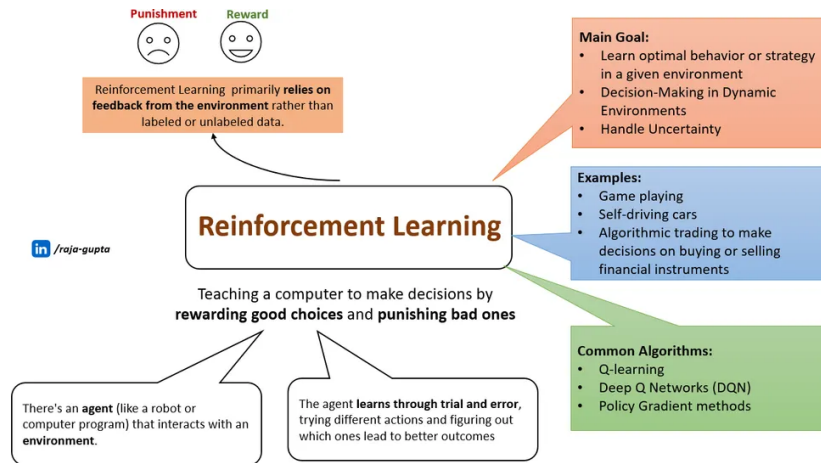


Figure 1.6: Reinforcement Learning summarized

Chapter 2

Algorithms

2.1 Regression Methods

Regression is one of the fundamental tasks in supervised learning. Its goal is to predict a continuous-valued output based on one or more input variables. Formally, given a dataset of input–output pairs $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^d$ represents the input features and $y_i \in \mathbb{R}$ represents a continuous target variable, a regression model seeks to learn a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that approximates the underlying relationship between inputs and outputs.

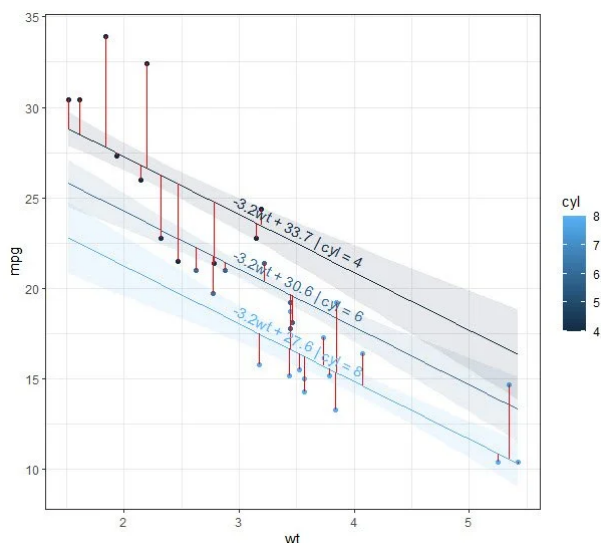


Figure 2.1: Regression examples

In a typical regression problem, the input can represent any structured or unstructured data — such as physical measurements, numerical attributes, or encoded representations of images, audio signals, or text. The model processes these inputs and outputs a single real number or, in the case of multivariate regression, a vector of continuous values.

The simplest and most widely known regression technique is *linear regression*, where the model assumes a linear relationship between the input features and the target:

$$\hat{y} = w^\top x + b,$$

where w is a weight vector and b is a bias term. More complex regression models, including polynomial regression, decision trees, and neural networks, can capture nonlinear dependencies between inputs and outputs.

The performance of a regression model is commonly evaluated using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), or Mean Absolute Error (MAE), which quantify how close the predicted values $\hat{y}_i$ are to the true targets y_i .

In summary, regression models map numerical representations of real-world phenomena to continuous outcomes, forming the basis for many predictive modeling tasks, from estimating housing prices and forecasting stock trends to predicting physical properties in scientific domains.

2.2 Classification Methods

Classification is another central task in supervised learning, where the objective is to assign each input to one of several discrete categories. Formally, given a dataset of labeled examples $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^d$ represents the feature vector and $y_i \in \{1, \dots, K\}$ denotes the class label, a classification model learns a function $f : \mathbb{R}^d \rightarrow \{1, \dots, K\}$ or, equivalently, a probability distribution over classes $P(y|x)$.

The model ingests an input vector—which may represent text, an image, an audio signal, or any numerical encoding—and outputs either a class label (hard classification) or a vector of probabilities indicating the model’s confidence for each class (soft classification). The most common approach for probabilistic outputs is to apply a softmax transformation to the model’s final layer, ensuring that all predicted probabilities sum to one.

Several families of models can perform classification:

- **Linear Models:** Logistic Regression and Linear Discriminant Analysis (LDA) are among the simplest and most interpretable classifiers. They model class boundaries using linear decision surfaces and can provide calibrated probability estimates.
- **Support Vector Machines (SVM):** These models seek an optimal separating hyperplane that maximizes the margin between classes. Kernelized SVMs can capture nonlinear decision boundaries.
- **Tree-Based Models:** Decision Trees, Random Forests, and Gradient Boosting Machines (such as XGBoost or LightGBM) partition the feature space hierarchically. They often yield strong predictive performance and can handle heterogeneous data types.
- **Probabilistic Models:** Naïve Bayes and Gaussian Mixture Models (GMMs) represent classes using explicit probability distributions, allowing inference via Bayes’ rule.
- **Neural Network Classifiers:** Deep architectures such as Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), and Transformers are capable of learning highly nonlinear decision functions directly from raw data.

The quality of a classification model is typically assessed through metrics such as accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (ROC–AUC). These measures quantify how effectively the model distinguishes among different classes and are essential for evaluating performance, particularly in the presence of class imbalance.

In summary, classification methods aim to map structured or unstructured input data to discrete output categories. From simple linear models to large-scale deep networks, the central challenge remains the same: to generalize from observed examples to unseen data, capturing the underlying structure that defines each class.

2.3 Clustering

Clustering is a fundamental technique in *unsupervised learning*. Its purpose is to discover inherent groupings within a dataset by identifying patterns or similarities among unlabeled examples. Unlike classification, where each data point is associated with a predefined label, clustering attempts to infer structure directly from the data itself. In this sense, it can be viewed as a way to explore the natural organization of data without prior supervision.

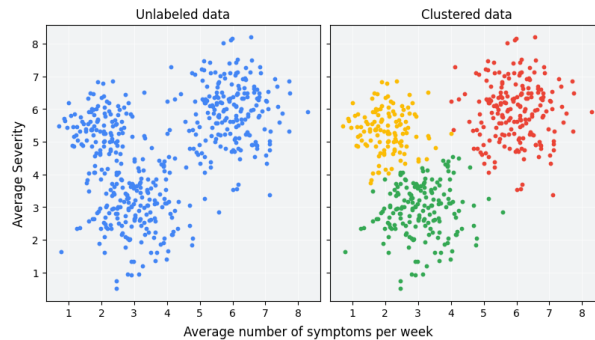


Figure 2.2: Unlabeled data vs Clustered data

Formally, given a set of examples $\{x_i\}_{i=1}^N$ with $x_i \in \mathbb{R}^d$, a clustering algorithm aims to partition the dataset into K groups, or *clusters*, such that data points within the same cluster are more similar to each other than to those in different clusters. Similarity is typically measured through distance metrics such as Euclidean, Manhattan, or cosine distance, depending on the data representation and problem context.

To illustrate the concept, consider a clinical study designed to evaluate a new treatment protocol. During the experiment, patients report both the frequency and severity of their symptoms per week. Since these observations are continuous and not labeled, researchers can apply clustering analysis to group patients with similar responses to the treatment. For example, one cluster may contain patients who show rapid improvement, another may represent those with moderate progress, and a third may include patients with minimal or no response.

Figure ?? shows a simulated example in which patient data are grouped into three distinct clusters based on their symptom profiles. Such analyses allow researchers to uncover hidden patterns in clinical outcomes and to tailor future interventions according to the characteristics of each patient subgroup.

Clustering Methods

A variety of clustering algorithms have been developed, each suited to different data structures and assumptions:

- **K-Means Clustering:** A centroid-based algorithm that partitions data into K clusters by minimizing the within-cluster sum of squared distances. It is simple and efficient, but assumes spherical cluster shapes and requires the number of clusters to be specified in advance.
- **Hierarchical Clustering:** Builds a nested hierarchy of clusters either by iteratively merging smaller clusters (agglomerative) or by splitting larger ones (divisive). The results are often visualized through a dendrogram, which provides insights into the data's structure at multiple levels of granularity.
- **Density-Based Clustering (DBSCAN, HDBSCAN):** Groups data points based on regions of high density, allowing the discovery of arbitrarily shaped clusters and the identification of noise or outliers. It does not require specifying the number of clusters a priori.
- **Gaussian Mixture Models (GMM):** A probabilistic approach in which each cluster is represented as a Gaussian distribution. Unlike K-Means, GMM provides *soft assignments*, meaning that each data point can belong to multiple clusters with different probabilities.
- **Spectral Clustering:** Utilizes the eigenvalues of a similarity matrix to perform clustering in a lower-dimensional space, making it suitable for non-convex or complex data manifolds.

The choice of clustering method depends on the data distribution, dimensionality, and the desired interpretability of results. In practice, clustering is often used as an exploratory tool for data analysis, feature engineering, or as a preprocessing step in larger machine learning pipelines.

In summary, clustering serves as a powerful approach to reveal latent structures within unlabeled datasets. It is widely applied in fields such as bioinformatics, market segmentation, image analysis, and recommender systems, where uncovering hidden patterns can lead to more meaningful understanding and informed decision-making.

2.4 Dimensionality Reduction

Dimensionality reduction is a key technique in machine learning and data analysis that seeks to represent high-dimensional data in a lower-dimensional space while preserving as much relevant information as possible. It serves multiple purposes: improving computational efficiency, mitigating the curse of dimensionality, reducing noise, and enhancing interpretability or visualization of complex datasets.

Formally, given a dataset $\{x_i\}_{i=1}^N$ where each example $x_i \in \mathbb{R}^D$ is described by D features, the goal is to find a transformation function $f : \mathbb{R}^D \rightarrow \mathbb{R}^d$ with $d \ll D$ such that the lower-dimensional representation $z_i = f(x_i)$ retains the most informative characteristics

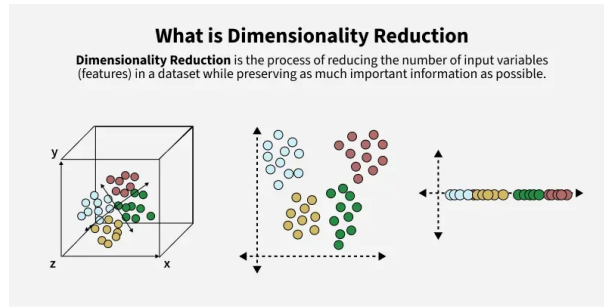


Figure 2.3: Dimensionality Reduction

of the original data. The challenge lies in compressing the representation without losing the essential structure or relationships between data points.

Dimensionality reduction techniques can be broadly divided into two categories: *feature selection* and *feature extraction*. Feature selection involves choosing a subset of the most relevant original variables, while feature extraction constructs new features as combinations or projections of the existing ones.

Several methods have been developed to perform dimensionality reduction, each based on different assumptions and optimization objectives:

- **Principal Component Analysis (PCA):** A linear technique that projects data onto the directions (principal components) of maximum variance. PCA identifies orthogonal axes that best explain the variability in the dataset and is often used for visualization and preprocessing.
- **Linear Discriminant Analysis (LDA):** Although originally designed for classification, LDA can also be used for dimensionality reduction by projecting data onto directions that maximize class separability.
- **Independent Component Analysis (ICA):** Aims to decompose multivariate data into statistically independent components, often used in signal processing and blind source separation tasks.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE):** A nonlinear technique that preserves local similarities among data points, making it particularly useful for visualizing high-dimensional data in two or three dimensions.
- **Uniform Manifold Approximation and Projection (UMAP):** A more recent nonlinear method that approximates the topological structure of data manifolds, often providing better scalability and global structure preservation than t-SNE.
- **Autoencoders:** Neural network-based models that learn a compressed latent representation of data by training to reconstruct the input from a lower-dimensional code. They can capture complex nonlinear relationships and are widely used in deep learning pipelines.

To illustrate, consider a dataset describing patients with hundreds of medical features, such as laboratory values and genetic markers. Applying PCA or UMAP can reveal low-dimensional embeddings that group patients with similar clinical profiles, enabling researchers to visualize disease subtypes or predict treatment outcomes more effectively.

In summary, dimensionality reduction transforms complex, high-dimensional datasets into compact and informative representations. It facilitates efficient learning, improved generalization, and deeper insight into the intrinsic structure of data—serving as both a practical preprocessing step and a powerful tool for exploratory data analysis.

2.5 Kernel Methods

Kernel methods provide a powerful framework for performing nonlinear learning using algorithms that are inherently linear in their formulation. The central idea is to map input data into a high-dimensional feature space where linear decision boundaries become sufficient for solving tasks that are nonlinear in the original space. This transformation is achieved implicitly through a *kernel function*, avoiding the computational cost of explicitly computing high-dimensional feature vectors.

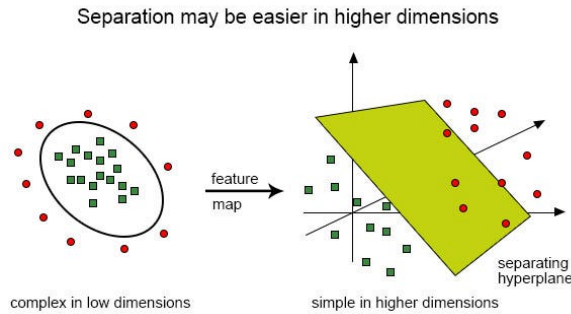


Figure 2.4: Kernel Method

2.5.1 Motivation and Intuition

Many datasets exhibit patterns that are not linearly separable. For example, consider points arranged in concentric circles: no straight line can separate them in the original input space. However, if we project the data into a higher-dimensional space—such as adding a radial distance feature—the classes may become linearly separable.

Kernel methods exploit this idea using the *kernel trick*: instead of explicitly computing

$$\phi(x) \in \mathcal{H},$$

a high-dimensional (possibly infinite-dimensional) feature representation, they compute only inner products in that space:

$$K(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}.$$

This allows algorithms like Support Vector Machines (SVMs) to operate efficiently while benefiting from the representational power of high-dimensional spaces.

2.5.2 The Kernel Trick

Many linear algorithms depend exclusively on inner products between data points. By replacing these inner products with a kernel function, the algorithm implicitly operates in feature space without ever computing $\phi(x)$ directly.

Formally, replace:

$$x^\top x' \longrightarrow K(x, x').$$

This technique enables:

- nonlinear classification (e.g., kernel SVMs),
- nonlinear regression (kernel ridge regression),
- dimensionality reduction (kernel PCA),
- clustering (spectral clustering).

The key advantage is that the dimensionality of $\phi(x)$ does not affect the computational complexity as long as $K(x, x')$ is efficient to compute.

2.5.3 Common Kernel Functions

Several kernels are widely used due to their theoretical properties and empirical success:

Linear Kernel

$$K(x, x') = x^\top x'.$$

Equivalent to no feature mapping. Useful as a baseline or when data are already separable with linear models.

Polynomial Kernel

$$K(x, x') = (x^\top x' + c)^p.$$

Allows the model to represent polynomial interactions of degree p . Good for tasks where feature interactions matter.

Radial Basis Function (RBF) Kernel

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right).$$

One of the most powerful and widely used kernels. Creates smooth decision boundaries and can represent extremely complex patterns.

Sigmoid (Neural) Kernel

$$K(x, x') = \tanh(\alpha x^\top x' + c).$$

Inspired by neural activation functions; historically linked to early neural network theory.

Custom or Domain-Specific Kernels String kernels, graph kernels, and convolution kernels allow the method to operate effectively on structured data, such as sequences or graphs.

2.5.4 Kernel SVM as a Primary Example

In Support Vector Machines, the classifier is expressed as:

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x, x_i) + b,$$

where only support vectors have nonzero coefficients α_i . Using a kernel replaces linear dot products, allowing the SVM to construct highly flexible nonlinear decision boundaries.

Intuition. Instead of fitting a straight line, kernel SVMs create curved boundaries tailored to the data geometry, while still maximizing the margin in feature space.

2.5.5 Advantages and Limitations

Advantages:

- Effective at capturing nonlinear patterns without explicit feature engineering.
- Strong theoretical background (reproducing kernel Hilbert spaces).
- Work well with smaller to medium-sized datasets.
- Highly flexible due to choice of kernel functions.

Limitations:

- Computational cost scales poorly with dataset size ($O(n^2)$ memory, $O(n^3)$ training).
- Hard to tune (choice of kernel + hyperparameters).
- Less competitive on extremely high-dimensional or massive datasets compared to deep learning.

2.5.6 Kernel Methods in Modern Machine Learning

While deep learning dominates large-scale tasks, kernel methods remain highly competitive and widely used in:

- bioinformatics (sequence kernels),
- small/tabular datasets,
- anomaly detection (one-class SVM),

- dimensionality reduction (kernel PCA),
- graph and structured data domains.

Hybrid models combining neural networks and kernels (e.g., deep kernel learning) are an active research area, leveraging the advantages of both worlds.

2.5.7 Summary

Kernel methods enable powerful nonlinear modeling through elegant mathematical principles and efficient implicit feature mappings. They form the backbone of many classical machine learning algorithms and remain an essential conceptual bridge between linear models and modern deep architectures. Understanding kernels provides insight into similarity measures, geometry of data, and high-dimensional representations—concepts that continue to influence contemporary machine learning.

Chapter 3

Models

3.1 Neural Network Architectures

Neural network architectures define the structural organization and information flow within models that learn to approximate complex functions. They specify how neurons are arranged into layers, how these layers are connected, and how data propagate through the network during training and inference. Each architecture introduces specific inductive biases that make it particularly suited for certain types of data or tasks.

At their core, neural networks consist of a series of transformations applied to an input vector $x \in \mathbb{R}^d$. Each layer performs a linear mapping followed by a nonlinear activation function:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}),$$

where $W^{(l)}$ and $b^{(l)}$ are the layer's trainable parameters, $\sigma(\cdot)$ is a nonlinear activation function such as ReLU or GeLU, and $h^{(l)}$ represents the layer's output (often referred to as the hidden representation). The final layer produces an output $\hat{y}$ that corresponds to a prediction or learned embedding, depending on the task.

The design of a neural network architecture determines:

- **Depth:** the number of layers stacked sequentially, which influences the level of abstraction the model can learn.
- **Width:** the number of neurons per layer, affecting the model's capacity and expressiveness.
- **Connectivity:** how neurons or layers are connected—whether fully, locally, recurrently, or via attention mechanisms.
- **Parameter sharing and constraints:** architectural choices such as convolutions or recurrent weights that impose structure and reduce the number of learnable parameters.

The flexibility of neural networks lies in their ability to adapt architectural principles to specific problem domains. For instance, some architectures emphasize spatial feature extraction, others model temporal dependencies, and others learn global contextual relationships. This diversity allows neural networks to serve as a unifying framework across computer vision, natural language processing, speech recognition, and many other fields.

In practice, the choice of architecture depends on the nature of the input data and the desired output:

- For structured, tabular, or vector data, fully connected or feedforward networks are often employed.
- For grid-like data such as images, convolutional architectures are preferred.
- For sequential data such as time series or text, recurrent or attention-based architectures are more effective.

Although these categories differ in structure and function, they share common building blocks: linear transformations, nonlinear activations, normalization mechanisms, and regularization techniques. The interplay of these components defines the representational power and generalization ability of the network.

In summary, neural network architectures represent the blueprint of how information is processed and transformed within a model. By designing the right architecture, one can embed domain knowledge directly into the network’s structure—guiding it to learn meaningful and efficient representations of data. The following sections present the most prominent architectural families—Feedforward, Convolutional, Recurrent, and Transformer networks—each tailored to specific data modalities and learning paradigms.

3.2 Feedforward Networks such as Multi Layer Perceptron (MLP)

Feedforward Neural Networks, commonly referred to as *Multilayer Perceptrons (MLPs)*, represent the simplest and most fundamental class of neural network architectures. They serve as the conceptual foundation upon which more advanced architectures—such as Convolutional, Recurrent, and Transformer models—are built. An MLP processes information strictly in one direction: from input to output, without any recurrent or feedback connections.

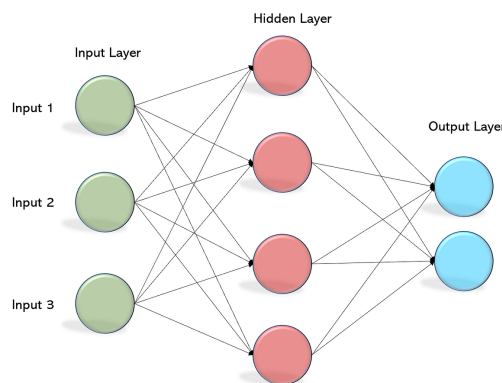


Figure 3.1: MLP

3.2.1 Structure and Forward Propagation

An MLP is composed of three main components:

- **Input layer:** receives the raw data, where each neuron corresponds to one input feature.
- **Hidden layers:** perform a sequence of linear and nonlinear transformations to extract intermediate representations.
- **Output layer:** produces the final prediction, whose dimensionality depends on the task (e.g., a single neuron for regression, or multiple neurons for classification).

Each layer in the network is fully connected to the next, meaning that every neuron in one layer influences every neuron in the subsequent layer through weighted connections. Information flows forward through the network, with no cycles or memory of previous inputs.

The data propagate through the network according to the following computation:

$$h^{(1)} = \sigma(W^{(1)}x + b^{(1)}), \quad h^{(2)} = \sigma(W^{(2)}h^{(1)} + b^{(2)}), \quad \hat{y} = f(W^{(3)}h^{(2)} + b^{(3)}),$$

where x represents the input vector, $W^{(l)}$ and $b^{(l)}$ are the trainable weights and biases of layer l , $\sigma(\cdot)$ is a nonlinear activation function (such as ReLU or GeLU), and $\hat{y}$ is the final network output.

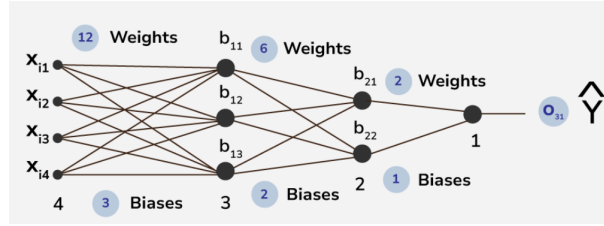


Figure 3.2: Forward Propagation

The use of nonlinear activation functions between layers allows MLPs to model complex, non-linear relationships between inputs and outputs. Without these nonlinearities, the composition of multiple layers would collapse into a single linear transformation, greatly limiting the model's expressive power.

3.2.2 Mathematical Formulation

Formally, a feedforward network defines a parameterized function $f_{\theta}(x)$ that maps an input $x \in \mathbb{R}^d$ to an output $\hat{y} \in \mathbb{R}^k$:

$$f_{\theta}(x) = W^{(L)} \sigma(W^{(L-1)} \sigma(\cdots \sigma(W^{(1)}x + b^{(1)})) + b^{(L-1)}) + b^{(L)}.$$

Here, $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^L$ represents all learnable parameters of the network. The number of layers L determines the network's depth, while the number of neurons per layer determines its width and representational capacity.

The *Universal Approximation Theorem* states that a feedforward network with at least one hidden layer and a non-linear activation function can approximate any continuous function on a compact subset of $\mathbb{R}^n$, given sufficient neurons. This theoretical result underscores the expressive power of even simple MLPs.

3.2.3 Training Process

Training an MLP involves adjusting its parameters θ to minimize a predefined loss function that quantifies the discrepancy between predicted outputs $\hat{y}$ and true targets y . This optimization is typically performed using stochastic gradient descent (SGD) or one of its adaptive variants such as Adam or RMSProp. Gradients of the loss function with respect to each parameter are computed efficiently using the *backpropagation* algorithm.

During training, techniques such as *Dropout*, *Batch Normalization*, and *Weight Regularization* are commonly applied to improve generalization and prevent overfitting. The learning rate, batch size, and number of hidden layers are hyperparameters that must be tuned based on the complexity of the dataset and the desired performance.

3.2.4 Applications and Limitations

Feedforward networks are versatile and can be applied to a wide range of problems, including:

- **Regression tasks:** predicting continuous outcomes such as prices, temperatures, or sensor readings.
- **Classification tasks:** assigning input samples to discrete categories.
- **Representation learning:** serving as encoders or decoders in larger architectures such as autoencoders.

However, MLPs treat all input features as independent and lack mechanisms to exploit structural information such as spatial or temporal relationships. This limitation motivates the development of specialized architectures like Convolutional Neural Networks (CNNs) for spatial data and Recurrent Neural Networks (RNNs) for sequential data.

In summary, Feedforward Networks and MLPs form the conceptual foundation of modern deep learning. They introduce the principles of layered computation, parameter optimization, and nonlinear transformation that underpin all subsequent neural architectures. Although simple in structure, they remain powerful tools for modeling generic data and continue to play a central role in hybrid and advanced neural systems.

3.3 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) represent a specialized class of neural architectures designed to process data with a grid-like topology, such as images, audio spectrograms, or video frames. They extend the concept of feedforward networks by introducing spatially local connections and shared weights, allowing them to efficiently capture hierarchical and translationally invariant features within structured inputs.

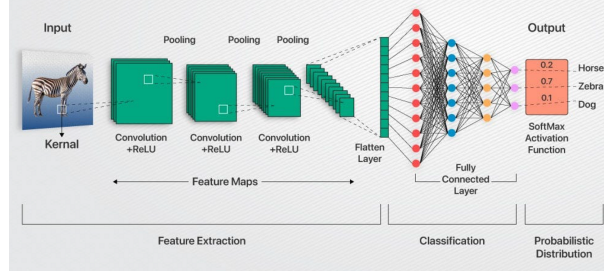


Figure 3.3: CNN

3.3.1 Motivation and Overview

Traditional feedforward networks treat all input features as independent and require a large number of parameters to process high-dimensional data, such as pixels in an image. CNNs address this limitation by exploiting the spatial correlation between neighboring input elements. Instead of connecting each neuron to every input unit, convolutional layers connect each neuron only to a small region of the input—called the *receptive field*.

This localized connectivity enables CNNs to detect low-level features (e.g., edges, corners, or textures) in early layers, and progressively more complex patterns (e.g., shapes or objects) in deeper layers. As a result, CNNs can learn compact and meaningful feature hierarchies with far fewer parameters than fully connected networks.

3.3.2 Convolutional Operations

The core operation of a CNN is the *convolution*, which applies a set of learnable filters (kernels) to the input data. For a 2D input image I and a filter K of size $m \times n$, the convolution operation is defined as:

$$S(i, j) = (I * K)(i, j) = \sum_{u=0}^{m-1} \sum_{v=0}^{n-1} I(i+u, j+v) K(u, v),$$

where $S(i, j)$ denotes the output feature map (also known as an activation map). Each filter extracts a specific type of feature, and multiple filters operating in parallel produce several feature maps that capture different aspects of the input.

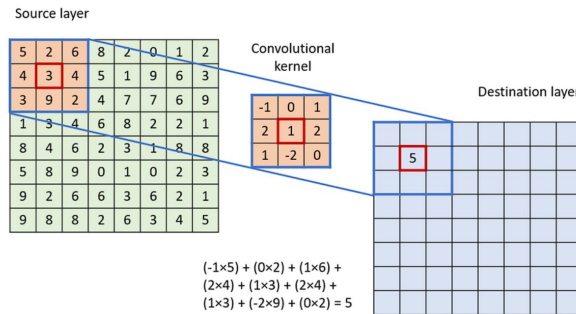


Figure 3.4: Convolution Illustration

After each convolutional layer, nonlinear activation functions (such as ReLU) introduce nonlinearity, enabling the network to model complex relationships between features.

These layers are often followed by pooling operations that downsample the spatial dimensions, reducing computational cost and increasing translational robustness.

3.3.3 Architectural Components

A typical CNN architecture consists of several types of layers arranged in sequence:

- **Convolutional layers:** perform feature extraction using multiple filters. The output of each convolution represents the response of a particular feature detector applied across the spatial domain.
- **Pooling layers:** reduce the spatial dimensions of feature maps by summarizing local regions (e.g., using max or average pooling), improving computational efficiency and spatial invariance.
- **Normalization layers:** such as Batch Normalization, stabilize and accelerate training by standardizing intermediate activations.
- **Fully connected layers:** often used at the end of the network to combine the extracted features into a final decision (e.g., class probabilities).

Early CNN architectures such as *LeNet-5* demonstrated the viability of this approach for digit recognition, while later models such as *AlexNet*, *VGGNet*, and *GoogLeNet* introduced deeper hierarchies and more efficient filter designs. Modern CNNs like *ResNet* employ *residual connections*—direct skip links between layers—to facilitate gradient flow and enable the training of extremely deep networks without performance degradation.

3.3.4 Training and Optimization

Training a CNN follows the same principles as other neural networks: parameters are optimized to minimize a loss function (e.g., cross-entropy for classification) using stochastic gradient descent (SGD) or adaptive optimizers such as Adam. However, CNN training introduces specific challenges related to overfitting, computational cost, and vanishing gradients.

To mitigate these issues, practitioners commonly employ:

- **Data augmentation:** artificially increasing the diversity of training data through transformations such as rotation, scaling, or flipping.
- **Regularization techniques:** such as Dropout and Weight Decay to improve generalization.
- **Batch Normalization:** to stabilize learning dynamics and accelerate convergence.
- **Transfer learning:** reusing pretrained CNNs (e.g., ResNet, VGG) as feature extractors for new tasks, significantly reducing training time and required data.

3.3.5 Applications and Impact

CNNs have become the cornerstone of modern computer vision and beyond. Their ability to learn spatial hierarchies of features has led to breakthroughs in tasks such as:

- Image classification and object detection (e.g., ImageNet, YOLO, Faster R-CNN)
- Semantic and instance segmentation (e.g., U-Net, Mask R-CNN)
- Image generation and restoration (e.g., autoencoders, GANs)
- Medical imaging, remote sensing, and visual inspection

Beyond vision, CNNs have also been successfully adapted to non-visual domains such as audio analysis, natural language processing, and time-series forecasting, where local dependencies and translation-invariant features are relevant.

In summary, Convolutional Neural Networks introduced the concept of local receptive fields and parameter sharing, revolutionizing how deep models handle structured data. Their hierarchical feature extraction and scalability make them one of the most influential architectures in the history of machine learning, forming the basis for numerous modern deep learning systems.

3.3.6 One-Dimensional Convolutional Neural Networks (1D-CNN)

While Convolutional Neural Networks (CNNs) are commonly associated with two-dimensional data such as images, the same principles can be extended to one-dimensional signals. *One-Dimensional Convolutional Neural Networks (1D-CNNs)* are specialized architectures designed to process sequential or temporal data, where information is arranged along a single spatial or time axis.

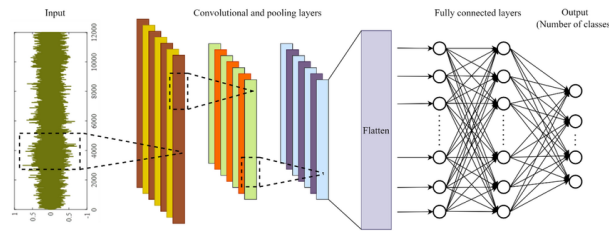


Figure 3.5: CNN-1d

In a 1D convolution, each filter slides over the input sequence along one dimension—typically time or ordered features—and performs a convolution operation defined as:

$$y[i] = \sum_{k=0}^{K-1} x[i+k] w[k],$$

where x is the input sequence, w is a kernel of size K , and y is the resulting feature map. Unlike 2D convolutions, which operate over height and width, 1D convolutions only consider local neighborhoods within the temporal or feature dimension.

This operation allows the network to capture *local dependencies* within a sequence—such as short-term temporal correlations or characteristic patterns in signal segments. By

stacking multiple convolutional and pooling layers, 1D-CNNs can progressively model more abstract and long-range dependencies, similar to how 2D CNNs capture hierarchical visual features.

Typical Architecture. A 1D-CNN typically includes:

- One or more **Conv1D layers**, each applying several filters to detect local patterns.
- **Pooling layers** (e.g., MaxPooling1D) that downsample the sequence, reducing its length while preserving salient features.
- Optionally, **Dropout layers** for regularization and **Batch Normalization** for training stability.
- One or more **fully connected layers** that map the extracted temporal features to the final output (e.g., class probabilities or regression outputs).

For example, a Conv1D layer processing an input of shape (batch, T, C) , where T is the sequence length and C is the number of channels or features per time step, produces an output feature map of shape (batch, T', F) —with F filters and a reduced temporal dimension T' determined by the stride and kernel size.

Applications. 1D-CNNs are widely applied in domains where data are naturally represented as ordered sequences, including:

- **Signal processing:** ECG, EMG, EEG, and other biomedical signal classification or anomaly detection.
- **Audio analysis:** speech command recognition, sound event detection, and waveform analysis.
- **Time-series forecasting:** financial, meteorological, or industrial sensor data.
- **Natural language processing:** sentence classification or keyword detection using fixed-length embeddings.

Compared to Recurrent Neural Networks (RNNs), 1D-CNNs can process sequences in parallel and often achieve faster training, while still capturing essential local temporal patterns through their convolutional filters. However, they are less effective in modeling very long-term dependencies, where recurrent or attention-based models (such as LSTMs or Transformers) typically perform better.

Summary. 1D-CNNs extend the power of convolutional architectures to one-dimensional structured data, combining efficiency, parallelism, and effective feature extraction for sequential tasks. Their balance between simplicity and expressiveness makes them a strong choice for many practical applications involving time-dependent or ordered input data.

3.4 Recurrent and Sequential Models (RNN, LSTM, GRU)

Recurrent and sequential neural architectures are specifically designed to model data where the ordering of elements carries essential meaning. Unlike feedforward and convolutional networks—which process inputs in fixed-size blocks—Recurrent Neural Networks (RNNs) operate on sequences of arbitrary length by maintaining an internal hidden state that evolves over time. This recurrent mechanism enables the network to capture temporal dependencies, making it suitable for tasks involving language, speech, time-series data, and any domain where context and order matter.

3.4.1 Motivation and Overview

Sequential data exhibit dependencies between elements: a word depends on the previous words in a sentence, a sensor reading depends on earlier readings, and the pitch of a sound varies continuously over time. Standard neural architectures fail to naturally capture these relationships because they treat each input independently.

Recurrent models address this limitation by incorporating a *memory mechanism*. At each time step t , the model receives an input x_t , updates its hidden state h_t , and optionally produces an output y_t . This process can be viewed as iteratively applying the same function across the sequence:

$$h_t = f(h_{t-1}, x_t),$$

where h_t stores information extracted from all previous inputs. This “shared-weight” structure allows RNNs to generalize across sequences of varying lengths.

3.4.2 Vanilla Recurrent Neural Networks (RNN)

The simplest recurrent model updates its hidden state using:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b),$$

and may produce an output through:

$$y_t = W_y h_t + c.$$

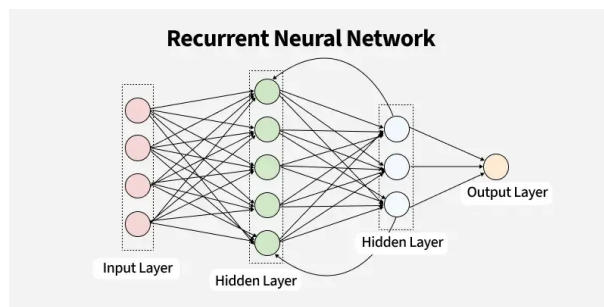


Figure 3.6: RNN

Strengths:

- Compact and computationally efficient.
- Able to capture short-term dependencies within sequences.

Limitations:

- **Vanishing gradients:** Gradients diminish during backpropagation through time, preventing the model from learning long-term dependencies.
- **Exploding gradients:** Opposite issue where gradients grow uncontrollably, often requiring gradient clipping.

Despite these limitations, vanilla RNNs form the conceptual basis for more advanced sequential architectures.

3.4.3 Long Short-Term Memory Networks (LSTM)

LSTM networks address the vanishing-gradient problem by introducing a more sophisticated memory cell with gating mechanisms. An LSTM maintains two states:

$$h_t \quad (\text{hidden state}), \quad c_t \quad (\text{cell state}),$$

where the cell state acts as a regulated memory highway, enabling the preservation of information over many time steps.

The LSTM performs the following gated operations:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) && (\text{forget gate}), \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) && (\text{input gate}), \\ \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) && (\text{candidate memory}), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t && (\text{updated cell state}), \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) && (\text{output gate}), \\ h_t &= o_t \odot \tanh(c_t). && (\text{updated hidden state}) \end{aligned}$$

LONG SHORT-TERM MEMORY NEURAL NETWORKS

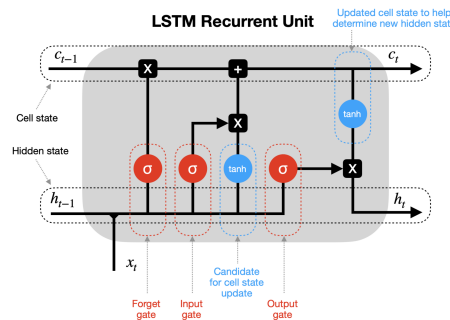


Figure 3.7: LSTM

Why LSTMs work:

- The forget gate f_t controls how much past information to retain.
- The input gate i_t controls how much new information to write.
- The output gate o_t controls how much of the memory to expose.

This flexible gating mechanism allows LSTMs to capture long-range dependencies and avoid the shortcomings of vanilla RNNs.

3.4.4 Gated Recurrent Units (GRU)

GRUs simplify the LSTM architecture while retaining its ability to handle long-term dependencies. They merge the cell and hidden states into a single state h_t and use only two gates:

$$\begin{aligned} z_t &= \sigma(W_z[h_{t-1}, x_t] + b_z) && \text{(update gate),} \\ r_t &= \sigma(W_r[h_{t-1}, x_t] + b_r) && \text{(reset gate),} \\ \tilde{h}_t &= \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h), \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \end{aligned}$$

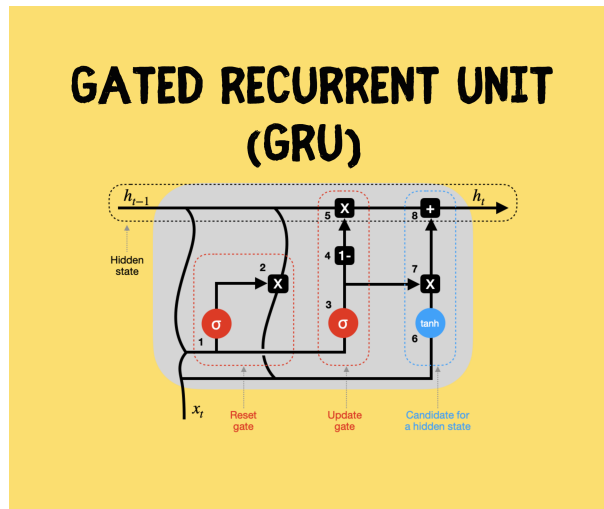


Figure 3.8: GRU

Advantages of GRUs:

- Fewer parameters than LSTMs $\rightarrow$ faster training and lower memory usage.
- Competitive or superior performance in many tasks.
- Simpler gating mechanism $\rightarrow$ easier to tune and analyze.

GRUs strike an effective balance between model expressiveness and computational efficiency.

3.4.5 Training and Practical Considerations

Backpropagation Through Time (BPTT). Training RNNs involves unrolling the network across time steps and computing gradients through each step. Longer sequences increase computational cost and exacerbate vanishing/exploding gradients.

Regularization. Sequential models often require:

- Dropout (applied carefully, e.g., variational dropout),
- Gradient clipping to prevent exploding gradients,
- Layer normalization for stability,
- Truncated BPTT for long sequences.

Comparison.

- RNNs: simplest, good for short dependencies.
- LSTMs: strong long-term memory, robust but computationally heavy.
- GRUs: lighter than LSTMs, often similar performance.

3.4.6 Applications

Recurrent models remain widely used in:

- Natural language processing (text generation, sentiment analysis),
- Speech recognition and audio modeling,
- Time-series forecasting (financial, environmental, industrial data),
- Sequential classification and anomaly detection.

Although many modern architectures (e.g., Transformers) outperform RNN-based models in large-scale NLP, recurrent models remain valuable for smaller datasets, real-time systems, and tasks requiring compact temporal reasoning.

3.4.7 Summary

Recurrent and sequential models introduce the ability to learn from ordered data via hidden-state dynamics. RNNs establish the foundation, LSTMs address long-term dependency issues through gating mechanisms, and GRUs offer a streamlined alternative with competitive performance. Understanding these architectures is essential for modeling any domain where temporal structure or sequential relationships are fundamental.

3.5 Transformer Architectures and Attention Mechanisms

Transformer architectures represent a fundamental shift in how neural networks process sequential data. Unlike recurrent models, which rely on recursive hidden-state updates, Transformers operate entirely through attention mechanisms—allowing them to capture global dependencies in parallel and with remarkable efficiency. This architecture has become the foundation of modern deep learning systems across natural language processing, computer vision, speech, and multimodal domains.

3.5.1 Motivation and Overview

Sequential models such as RNNs and LSTMs process data token by token, limiting parallelization and making it difficult to model long-range dependencies due to gradient degradation. Transformers address these limitations by replacing recurrence with a *self-attention* mechanism that directly relates every element of the input sequence to every other element, regardless of distance.

The core idea is to learn *contextualized representations* of each token by considering its interactions with all other tokens in the sequence. This approach allows the model to:

- process all sequence positions in parallel,
- model long-term relationships with ease,
- scale efficiently with modern hardware.

These properties have led Transformers to dominate contemporary architectures, particularly in large-scale language models such as BERT, GPT, T5, and multimodal systems like Vision Transformers (ViT) and CLIP.

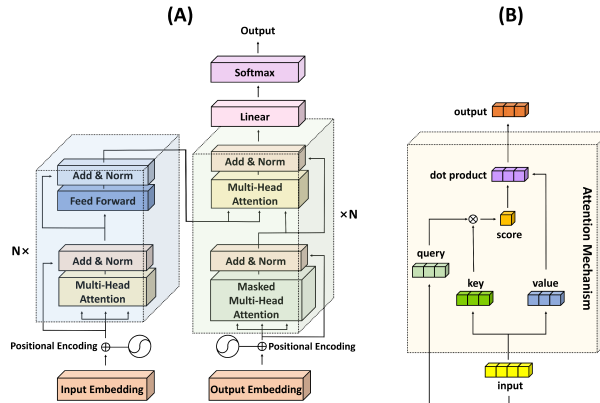


Figure 3.9: Transformer Architectures and Attention Mechanisms

3.5.2 Self-Attention Mechanism

The self-attention mechanism computes the relevance of each token to every other token. Given an input sequence represented as embeddings $X \in \mathbb{R}^{T \times d}$ (length T , dimension d),

three learned linear projections produce:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V,$$

where Q , K , and V denote the queries, keys, and values.

Self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where d_k is the dimensionality of the keys. The softmax term produces an attention weight matrix that determines how much each token contributes to the representation of every other token.

Intuition:

- Each query vector asks: “What other parts of the sequence are relevant to me?”
- Keys provide information for matching queries.
- Values supply the content to be aggregated based on relevance.

This mechanism allows the network to build flexible, context-aware representations of the entire sequence.

3.5.3 Multi-Head Attention

Instead of using a single attention computation, Transformers employ *multi-head attention*, where multiple attention heads operate in parallel, each learning a distinct type of relationship.

For H heads:

$$\text{MHA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W^O,$$

where each head performs:

$$\text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V).$$

Why multiple heads?

- Capture diverse patterns (syntax, semantics, global vs. local features).
- Stabilize learning by distributing representational burden.
- Improve expressiveness and flexibility.

3.5.4 Positional Encoding

Because Transformers lack recurrence and convolution, they require an explicit mechanism to encode sequence order. Positional encodings inject information about token positions into the embeddings:

$$PE_{(t,2i)} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad PE_{(t,2i+1)} = \cos\left(\frac{t}{10000^{2i/d}}\right),$$

where t is the position index and i indexes embedding dimensions. Alternatively, learned positional embeddings can be used.

Purpose:

- Allow the model to distinguish token ordering.
- Provide periodic structure useful for capturing relative distances.

3.5.5 Transformer Encoder and Decoder

The original Transformer architecture, introduced in “Attention Is All You Need,” consists of an *encoder* and *decoder*, each composed of repeated layers.

Encoder layer:

- Multi-Head Self-Attention
- Feedforward network (pointwise MLP)
- Layer normalization
- Residual connections

Decoder layer:

- Masked multi-head self-attention (prevents peeking at future tokens)
- Cross-attention over encoder outputs
- Feedforward network
- Layer normalization + residuals

This structure enables sequence-to-sequence tasks such as translation, summarization, and question answering.

3.5.6 Feedforward Networks in Transformers

Each Transformer layer includes a position-wise feedforward network:

$$\text{FFN}(x) = W_2 \sigma(W_1 x + b_1) + b_2,$$

often using the GeLU activation. These networks expand the dimensionality of hidden representations before compressing them back, increasing expressiveness within each layer.

3.5.7 Training, Optimization, and Scalability

Transformers train effectively using adaptive optimizers (e.g., AdamW) combined with learning rate warm-up and cosine decay schedules. Their key advantages include:

- Full parallelism across sequence positions,
- Ability to scale to billions of parameters,
- Efficient use of GPU/TPU hardware,
- Robust modeling of long-range dependencies.

However, self-attention has quadratic complexity $O(T^2)$ with respect to sequence length, motivating research into efficient alternatives such as Longformer, Performer, and FlashAttention.

3.5.8 Applications and Impact

Transformers have revolutionized deep learning and now dominate:

- Natural language processing (machine translation, large language models),
- Computer vision (Vision Transformers, segmentation, multimodal fusion),
- Speech processing and audio modeling,
- Time-series forecasting,
- Multi-modal and cross-modal systems (CLIP, Flamingo, GPT-4/5).

Their flexibility and scalability have made them the leading architecture for state-of-the-art models across domains.

3.5.9 Summary

Transformer architectures fundamentally changed sequential modeling by replacing recurrence with attention. Self-attention enables the network to directly capture global dependencies, while multi-head mechanisms enrich contextual understanding. Positional encodings restore sequence structure, and feedforward layers enhance representational power. These innovations have established Transformers as the core architecture driving modern AI systems.

Chapter 4

Core Components and Techniques

4.1 Activation Functions (ReLU, GeLU, Sigmoid, Tanh)

Activation functions are essential components of neural networks that introduce nonlinearity into the model. Without them, even a deep network composed of multiple layers would collapse into a single linear transformation, severely limiting its ability to approximate complex functions. By applying a nonlinear mapping to each neuron's output, activation functions enable networks to learn intricate relationships, extract abstract representations, and capture hierarchical patterns in data.

4.1.1 Theoretical Role of Activation Functions

In a neural layer, the pre-activation value for a neuron is computed as:

$$z = Wx + b,$$

where W and b are the layer's parameters, and x is the input vector. The activation function $\sigma(\cdot)$ then transforms z into the neuron's output:

$$h = \sigma(z).$$

This nonlinearity is crucial for enabling the network to represent functions that are not linearly separable. It allows stacked layers to progressively build complex mappings from simple compositions.

Different activation functions exhibit different behaviors in terms of gradient flow, computational efficiency, and representational capacity. The choice of activation can significantly impact convergence speed, model stability, and overall performance.

4.1.2 Rectified Linear Unit (ReLU)

The *Rectified Linear Unit* (*ReLU*) is one of the most commonly used activation functions in deep learning, defined as:

$$\text{ReLU}(x) = \max(0, x).$$

ReLU outputs the input directly if it is positive and zero otherwise, introducing sparsity in the activations and enabling efficient computation.

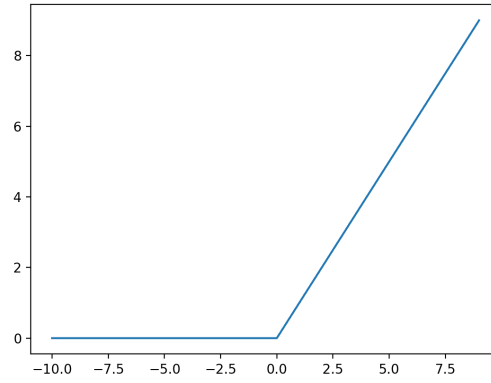


Figure 4.1: ReLU activation function

Advantages:

- Promotes sparse activation, reducing interdependence among neurons.
- Avoids gradient saturation for positive values, leading to faster convergence.
- Computationally simple and efficient.

Disadvantages:

- **Dying ReLU problem:** Neurons can become inactive (always output zero) when their inputs remain negative, effectively “dying” during training.

Example intuition: Consider a regression model predicting house prices. ReLU allows positive features like “number of rooms” or “square meters” to contribute linearly to the prediction, while naturally ignoring negative or irrelevant signals.

4.1.3 Gaussian Error Linear Unit (GeLU)

The *Gaussian Error Linear Unit (GeLU)* is a smoother alternative to ReLU, defined as:

$$\text{GeLU}(x) = x \Phi(x),$$

where $\Phi(x)$ is the cumulative distribution function (CDF) of the standard normal distribution. In practice, it is often approximated by:

$$\text{GeLU}(x) \approx 0.5x \left(1 + \tanh \left[\sqrt{\frac{2}{\pi}} (x + 0.044715x^3) \right] \right).$$

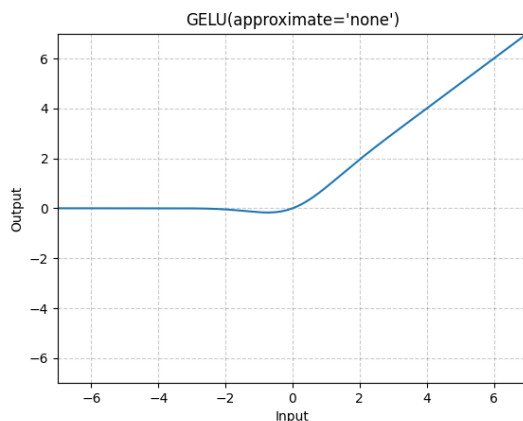


Figure 4.2: GELU activation function

Intuition: Instead of deterministically zeroing out negative values (like ReLU), GeLU smoothly scales them based on their magnitude—small negative inputs are partially suppressed, while larger positive ones pass almost unchanged. This stochastic gating behavior improves representational richness and gradient flow.

Advantages:

- Smooth and differentiable everywhere.
- Empirically improves performance in Transformer architectures (e.g., BERT, GPT).
- Retains some gradient for small negative inputs, mitigating neuron death.

When to use: Ideal for large-scale language or vision models where smoothness and stable gradient propagation are critical.

4.1.4 Sigmoid Function

The *Sigmoid* activation function maps any real-valued input into the range $(0, 1)$:

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

It is historically one of the first activation functions used in neural networks and remains common in the output layers of binary classification models.

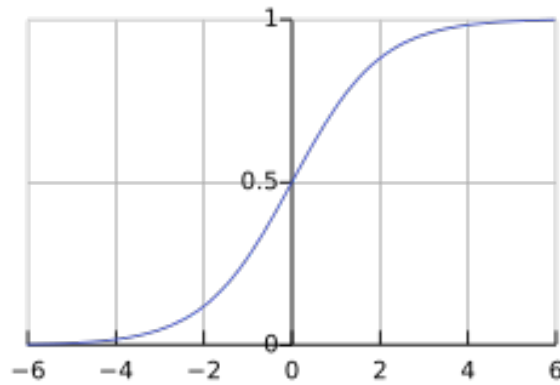


Figure 4.3: Sigmoid function activation

Advantages:

- Outputs interpretable as probabilities.
- Smooth and differentiable, suitable for probabilistic modeling.

Disadvantages:

- **Vanishing gradients:** For large positive or negative inputs, derivatives approach zero, slowing or halting learning.
- **Non-zero-centered output:** Causes gradient bias and slower convergence in deep networks.

Example: In binary classification (e.g., spam vs. non-spam), the sigmoid is often used in the output layer to predict the probability of belonging to the positive class.

4.1.5 Hyperbolic Tangent (Tanh)

The *Tanh* function is a rescaled version of the sigmoid:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

It maps inputs to the range $(-1, 1)$ and is zero-centered, which helps balance positive and negative activations.

Advantages:

- Zero-centered output helps with gradient-based optimization.
- Retains stronger gradients than sigmoid near the origin.

Disadvantages:

- Still prone to vanishing gradients for large $|x|$.
- More computationally expensive than ReLU.

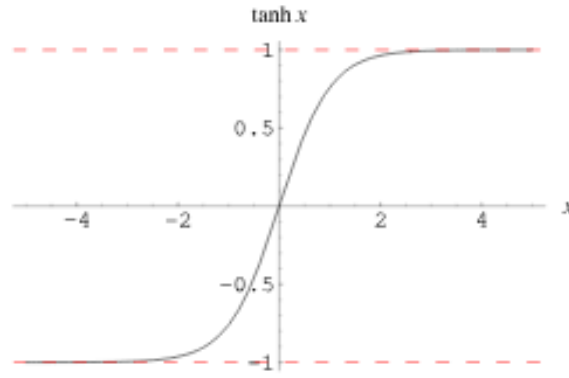


Figure 4.4: Hyperbolic Tangent activation function

Example: Tanh activations are often used in recurrent networks such as RNNs and LSTMs to model nonlinear dependencies while maintaining bounded activations.

4.1.6 Comparison and Practical Recommendations

- **ReLU:** Default choice for most networks due to simplicity and efficiency.
- **GeLU:** Preferred in modern deep architectures (Transformers) for smooth gradients and improved expressiveness.
- **Sigmoid:** Useful primarily for binary outputs or probabilistic interpretations.
- **Tanh:** Effective for sequential data and in hidden layers of recurrent architectures.

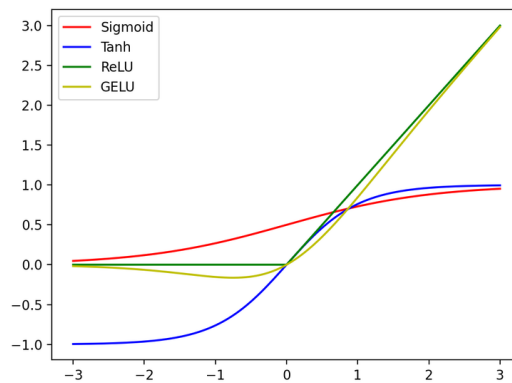


Figure 4.5: Comparison between Sigmoid, Tanh, ReLU and GeLU

In practice, activation choice affects not only convergence speed but also final model performance and stability. ReLU remains a strong baseline for most vision and feedforward models, while GeLU is dominant in NLP and Transformer-based systems. Sigmoid and Tanh retain specialized relevance in probabilistic and recurrent contexts.

4.1.7 Summary

Activation functions provide the nonlinear transformations that allow deep networks to model complex, high-dimensional relationships. While ReLU introduced simplicity and efficiency, newer functions such as GeLU refined smoothness and gradient behavior. The appropriate choice depends on the task, data distribution, and architecture—but understanding their mathematical properties remains central to effective neural network design.

4.2 Linear and Dense Layers (e.g., `nn.Linear`)

Linear or dense layers constitute one of the fundamental building blocks of neural networks. They apply an affine transformation to their input, enabling the model to learn weighted combinations of features. Despite their simplicity, linear layers are essential for constructing deep architectures, performing dimensionality transformations, and forming the core of many modern models—from feedforward networks to Transformers.

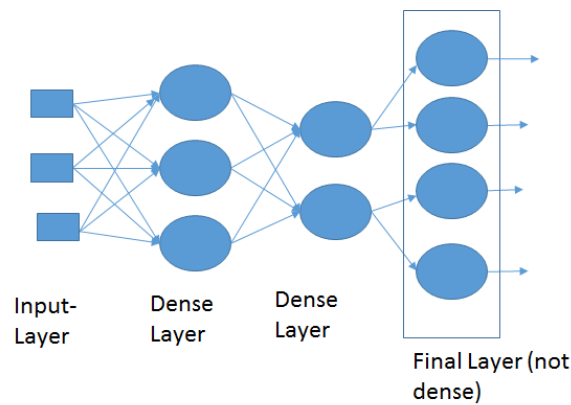


Figure 4.6: Linear and Dense Layers

4.2.1 Definition and Mathematical Formulation

A linear layer implements the mapping:

$$y = Wx + b,$$

where:

- $x \in \mathbb{R}^{d_{\text{in}}}$ is the input vector,
- $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is the weight matrix,
- $b \in \mathbb{R}^{d_{\text{out}}}$ is the bias vector,
- $y \in \mathbb{R}^{d_{\text{out}}}$ is the output.

In PyTorch, this operation is implemented via:

$$\text{nn.Linear}(d_{\text{in}}, d_{\text{out}}),$$

which automatically handles weight initialization, parameter storage, and gradient computation.

The layer conducts a weighted sum of the input components and adds a learnable bias term. Without any activation function, this transformation remains strictly linear; stacking several linear layers without nonlinearities collapses into an equivalent single linear transformation.

4.2.2 Role Within Neural Architectures

Linear layers serve various purposes across different architectures:

Feature Transformation. They project inputs into new feature spaces, often changing dimensionality (e.g., reducing or expanding representation size).

Combining Extracted Features. After convolutional or recurrent layers, linear layers aggregate learned features and prepare them for classification or regression tasks.

Final Output Layers. Many models conclude with a linear layer that maps internal representations to:

- probabilities (via a subsequent softmax),
- numerical predictions (regression),
- logits for classification.

Transformers and Attention. In Transformers, linear projections generate queries, keys, and values:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V,$$

and the feedforward sublayers rely entirely on dense transformations.

4.2.3 Dimensionality and Shape Considerations

Given an input tensor of shape:

$$(\text{batch}, T, d_{\text{in}}),$$

a linear layer applied along the last dimension returns:

$$(\text{batch}, T, d_{\text{out}}),$$

allowing seamless integration into networks handling sequences, time-series, and token embeddings.

In feedforward networks, the common shape is:

$$(\text{batch}, d_{\text{in}}) \rightarrow (\text{batch}, d_{\text{out}}).$$

The ability to operate on arbitrarily shaped tensors makes linear layers highly versatile.

4.2.4 Initialization and Trainability

Linear layers typically initialize weights using distributions designed to maintain stable gradients:

- Xavier/Glorot initialization for symmetric activations,
- Kaiming/He initialization for ReLU-based networks.

Poor initialization can hinder learning, cause vanishing or exploding gradients, or slow convergence. Proper initialization is especially important in deep networks where many linear layers are stacked sequentially.

4.2.5 Regularization and Stabilization Techniques

Dense layers often require additional components to ensure stable and generalizable training:

- **Dropout** to prevent co-adaptation and reduce overfitting,
- **BatchNorm** or **LayerNorm** to stabilize activations,
- **Weight decay** to constrain the magnitude of weights.

When combined with nonlinear activations (ReLU, GeLU), these techniques turn linear layers into powerful function approximators.

4.2.6 Example: A Simple Two-Layer MLP Block

Consider the block:

$$h = \sigma(W_1x + b_1), \quad y = W_2h + b_2,$$

where σ is a nonlinear activation function. This structure forms the foundational unit of a Multi-Layer Perceptron (MLP). The combination of two linear transformations with a nonlinearity allows the network to approximate highly complex functions.

4.2.7 Summary

Linear layers provide flexible and efficient mappings between feature spaces, forming the backbone of both classical and modern neural architectures. By learning weighted combinations of inputs, they enable deep networks to extract, transform, and combine representations across tasks and data modalities. Despite their simplicity, dense layers remain indispensable components of deep learning systems and interact closely with activation functions, normalization layers, and regularization techniques to build expressive models.

4.3 Normalization Layers (BatchNorm, LayerNorm, Group-Norm)

Normalization layers are essential components in modern neural networks, designed to stabilize training, improve convergence speed, and enhance generalization. They operate by normalizing activations according to specific statistical properties, reducing internal covariate shift and smoothing the optimization landscape. Different normalization strategies exist to accommodate various data modalities, batch sizes, and architectural constraints. This section covers three of the most widely used methods: Batch Normalization, Layer Normalization, and Group Normalization.

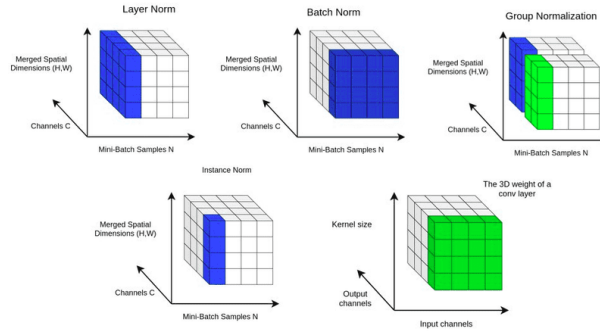


Figure 4.7: Normalization Technique

4.3.1 Motivation for Normalization

During training, the distribution of activations in intermediate layers can shift as parameters update—a phenomenon known as *internal covariate shift*. This shift makes optimization more difficult and often requires carefully tuned learning rates or initialization schemes.

Normalization layers mitigate this issue by:

- stabilizing the distribution of activations,
- smoothing the loss landscape,
- enabling deeper networks to train effectively,
- reducing sensitivity to initialization,
- acting as implicit regularizers.

While the core idea—subtracting a mean and scaling by a variance—is shared, each normalization method differs in how these statistics are computed.

4.3.2 Batch Normalization (BatchNorm)

Batch Normalization normalizes activations across the *batch dimension*. Given activations $x \in \mathbb{R}^{N \times C \times H \times W}$ (typical for CNNs), BatchNorm computes:

$$\mu_c = \frac{1}{N \cdot H \cdot W} \sum_{n,h,w} x_{nchw}, \quad \sigma_c^2 = \frac{1}{N \cdot H \cdot W} \sum_{n,h,w} (x_{nchw} - \mu_c)^2.$$

The normalized output is:

$$\hat{x}_{nchw} = \frac{x_{nchw} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}.$$

BatchNorm then applies learnable affine parameters:

$$y_{nchw} = \gamma_c \hat{x}_{nchw} + \beta_c.$$

Advantages:

- Strong empirical success in CNNs.
- Allows higher learning rates and faster convergence.
- Acts as a regularizer, often reducing the need for dropout.

Limitations:

- Performance degrades with small batch sizes.
- Unreliable in recurrent models due to temporal ordering.
- Requires maintaining running averages for inference.

4.3.3 Layer Normalization (LayerNorm)

Layer Normalization normalizes across the *feature dimension* rather than the batch dimension. Given activations $x \in \mathbb{R}^{N \times T \times D}$ (typical for sequences or token embeddings), LayerNorm computes:

$$\mu = \frac{1}{D} \sum_{i=1}^D x_i, \quad \sigma^2 = \frac{1}{D} \sum_{i=1}^D (x_i - \mu)^2.$$

Each sample is normalized independently:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta.$$

Advantages:

- Stable for small batch sizes (batch size 1 allowed).
- Highly effective for sequential models (RNNs, Transformers).
- Eliminates dependency on batch statistics.

Limitations:

- Slightly slower than BatchNorm in CNNs due to lack of batch parallelism.
- May not perform as well as BatchNorm in pure convolutional pipelines.

LayerNorm is the default normalization in Transformers, applied before or after attention and feedforward sublayers.

4.3.4 Group Normalization (GroupNorm)

Group Normalization divides the channels into groups and normalizes within each group. Given activations $x \in \mathbb{R}^{N \times C \times H \times W}$, with G groups, each group contains C/G channels. GroupNorm computes mean and variance for each group:

$$\mu_g = \frac{1}{m} \sum_{i \in \text{group } g} x_i, \quad \sigma_g^2 = \frac{1}{m} \sum_{i \in \text{group } g} (x_i - \mu_g)^2,$$

where m is the number of elements per group.

The normalized output is computed similarly to other methods, followed by learnable γ and β .

Advantages:

- Independent of batch size—excellent for small-batch or single-sample training.
- Works consistently for CNNs and dense architectures.
- A generalization: LayerNorm = GroupNorm with $G = C$, InstanceNorm = $G = C$ with no affine parameters.

Limitations:

- Requires choosing the number of groups.
- Slightly heavier computation than BatchNorm.

GroupNorm became useful in applications that rely on small batch sizes such as medical imaging, GAN training, or heavy data augmentation pipelines.

4.3.5 Comparison and Practical Guidelines

- **BatchNorm** is usually the best choice for CNNs when batch size is sufficiently large.
- **LayerNorm** is the preferred method for Transformers, RNNs, and any sequence-based model.
- **GroupNorm** is effective for convolutional architectures when batch sizes are small or unstable.

The choice of normalization significantly affects convergence behavior, stability, and final performance—especially in deep models.

4.3.6 Summary

Normalization layers improve training stability and efficiency by stabilizing activation distributions. BatchNorm excels in convolutional networks with large batches, LayerNorm dominates in sequence models and Transformers, and GroupNorm fills the gap when batch size constraints limit BatchNorm’s effectiveness. Together, these techniques play a crucial role in enabling deep networks to train reliably and generalize effectively across diverse data domains.

4.4 Dropout and Regularization Techniques

Regularization refers to a collection of strategies used to improve a model’s generalization ability and prevent overfitting. As neural networks increase in depth and capacity, they become prone to memorizing training data rather than learning patterns that generalize to unseen samples. Dropout and other regularization methods mitigate this risk by constraining the model, injecting noise, or penalizing overly complex solutions. This section provides a clear overview of the most commonly used regularization techniques in modern deep learning.

4.4.1 The Problem of Overfitting

Overfitting occurs when a model performs well on training data but poorly on validation or test data. It typically arises when:

- the model has too many parameters relative to dataset size,
- training is excessively long without proper constraints,
- noise or outliers fit “too tightly,”
- the model learns spurious correlations instead of true structure.

Regularization introduces bias or constraints that help guide the model toward simpler, more robust solutions.

4.4.2 Dropout

Dropout is one of the most widely used regularization techniques in deep neural networks. During training, it randomly “drops” a fraction of neuron activations by setting them to zero with probability p :

$$h'_i = \begin{cases} 0 & \text{with probability } p, \\ h_i/(1-p) & \text{with probability } 1-p. \end{cases}$$

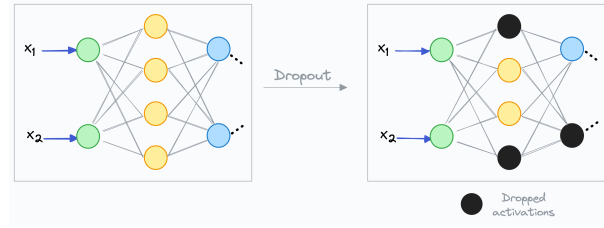


Figure 4.8: Dropout

The scaling factor $1/(1 - p)$ ensures that the expected activation magnitude remains consistent between training and inference.

Intuition. Dropout can be viewed as training an ensemble of networks that share weights:

- Each training pass samples a different subnetwork.
- No single neuron can rely on other specific neurons.
- The network learns redundant, robust representations.

Typical choices.

- $p = 0.5$ for fully connected layers,
- $p = 0.1\text{--}0.3$ for convolutional or recurrent architectures,
- specialized variants in RNNs ensure dropout is applied consistently across time steps.

Advantages.

- Strong reduction of co-adaptation between neurons.
- Effective in large MLPs, CNNs, and some RNN variants.
- Simple, efficient, and widely supported in all frameworks.

4.4.3 Weight Decay (L2 Regularization)

Weight decay penalizes large weights by adding an L_2 penalty to the loss:

$$\mathcal{L}'(\theta) = \mathcal{L}(\theta) + \lambda \|\theta\|_2^2.$$

This encourages the model to distribute importance across many weights rather than depending heavily on a few large parameters.

Benefits.

- Improves generalization by reducing weight magnitudes.
- Promotes smoother decision boundaries and less variance.

AdamW modifies this concept by decoupling weight decay from gradient-based updates—now standard in Transformer training.

4.4.4 Early Stopping

Early stopping halts training once validation performance stops improving. Even if training loss continues to decrease, the model may begin to overfit, so stopping at the optimal point ensures better generalization.

Procedure.

- Monitor validation loss or accuracy each epoch.
- Stop training after a patience window with no improvement.
- Restore the model to the best checkpoint encountered.

This technique is simple yet remarkably effective.

4.4.5 Data Augmentation

In tasks like image or audio classification, data augmentation artificially expands the training set by applying label-preserving transformations:

- random rotations, flips, crops,
- color jittering,
- time warping or frequency masking (audio),
- random erasing or cutout.

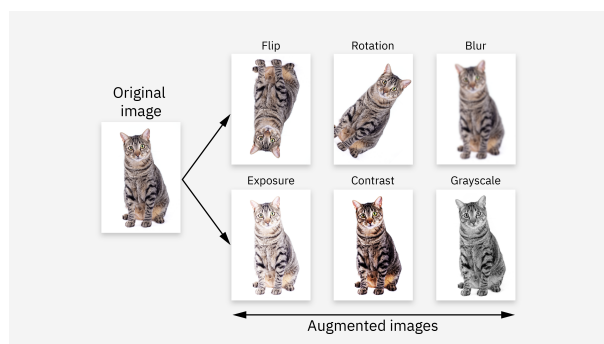


Figure 4.9: Data Augmentation

Augmentation reduces overfitting by increasing data diversity and forcing the model to learn invariances.

4.4.6 Label Smoothing

Label smoothing prevents the model from becoming overconfident by softening the target distribution. Instead of a one-hot label $y_k = 1$, we assign:

$$y_k = 1 - \epsilon, \quad y_{j \neq k} = \frac{\epsilon}{K - 1},$$

with smoothing parameter ϵ and number of classes K .



Figure 4.10: Label Smoothing

Effects.

- Prevents overconfident predictions.
- Improves calibration and robustness.
- Widely used in Transformers and classification models.

4.4.7 Other Regularization Strategies

Noise Injection. Add noise to inputs, hidden states, or weights to reduce sensitivity to small fluctuations.

DropConnect. Randomly drop weights instead of activations—similar spirit to dropout.

Stochastic Depth. Randomly skip entire layers during training; used in deep ResNets and Vision Transformers.

Mixup and CutMix. Blend or mix portions of training examples along with labels to improve robustness and reduce reliance on spurious correlations.

4.4.8 Choosing the Right Regularization Technique

- **Large MLPs:** Dropout + weight decay.
- **CNNs:** Data augmentation + weight decay; small dropout in dense layers.
- **Transformers:** Dropout + LayerNorm + AdamW + label smoothing.

- **RNNs:** Variational dropout + gradient clipping.

Specific architectures and datasets benefit from different forms of regularization, and combining several is often most effective.

4.4.9 Summary

Regularization techniques are crucial for ensuring that neural networks generalize beyond their training data. Dropout introduces stochastic sparsity to encourage redundancy, weight decay penalizes overly large parameters, early stopping prevents late-stage overfitting, and data augmentation expands the training manifold. Together, these methods stabilize optimization, improve robustness, and are essential tools for building reliable deep learning models.

4.5 Residual Connections and Skip Layers

Residual connections (or skip layers) are architectural mechanisms that enable deep neural networks to train more effectively by allowing information and gradients to bypass one or more layers. They address optimization difficulties that arise as networks grow deeper, such as vanishing gradients, slow convergence, and the degradation problem—where adding more layers results in higher training error.

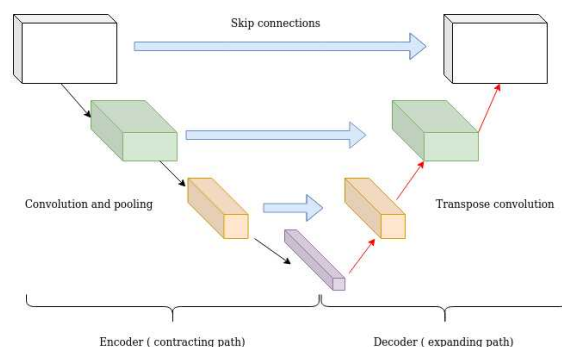


Figure 4.11: Residual Connections and Skip Layers

4.5.1 Core Idea

Given an input x and a transformation $F(x)$ (e.g., a sequence of convolutions, linear layers, or attention blocks), a residual connection outputs:

$$y = F(x) + x.$$

This formulation allows the network to learn *residual functions* rather than complete mappings, making identity transformations easy to represent and simplifying gradient flow.

4.5.2 Types of Skip Connections

Identity Skip. Used when input and output have the same shape:

$$y = F(x) + x.$$

Projection Skip. Used to match channel or dimensionality changes:

$$y = F(x) + W_s x.$$

Concatenation Skips. Combine features by concatenating:

$$y = \text{Concat}(x, F(x)),$$

used in U-Net and DenseNet to preserve spatial information and promote feature reuse.

Gated Skips. Learn a gate controlling the mixture between input and transformed features:

$$y = g \odot F(x) + (1 - g) \odot x.$$

4.5.3 Benefits

- Improve gradient propagation in very deep networks.
- Enable training of models with 100+ layers.
- Reduce sensitivity to initialization.
- Facilitate architectural modularity (e.g., residual blocks).
- Allow identity mappings when deeper transformations are unnecessary.

4.5.4 Use in Modern Architectures

Residual connections are foundational in:

- **ResNets:** convolutional residual blocks with identity or projection skips.
- **Transformers:** attention and feedforward blocks wrapped with residual paths.
- **U-Net:** long skip connections linking encoder and decoder stages.
- **Recurrent networks:** highway networks and gated residual architectures.

4.5.5 Summary

Skip and residual connections enable stable optimization, deeper architectures, and efficient information flow. They represent a fundamental design principle in modern deep learning, significantly improving trainability and performance across CNNs, Transformers, and many other architectures.

4.6 Weight Initialization and Parameterization

Weight initialization and parameterization play a crucial role in the stability, efficiency, and convergence of neural network training. Poor initialization can lead to exploding or vanishing activations, unstable gradients, and severely hinder the optimization process. Proper initialization ensures that signals propagate with the correct scale through the network, enabling deeper models to train effectively and reducing sensitivity to hyperparameters such as learning rate.

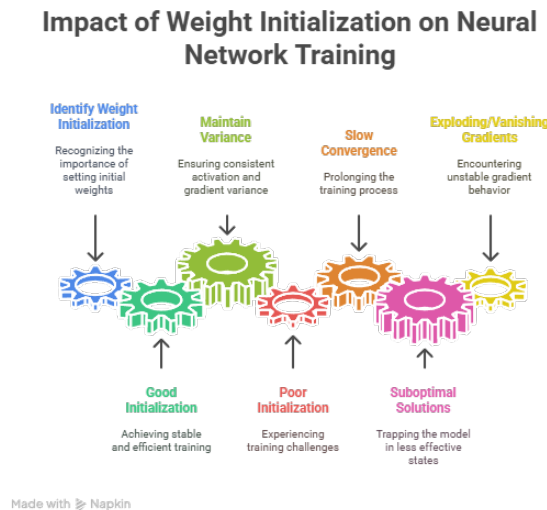


Figure 4.12: Difference between Good and Bad Initialization

4.6.1 The Role of Initialization

Neural networks rely on gradient-based optimization. If activations or gradients grow or shrink exponentially across layers, training becomes unstable. This phenomenon is most evident when:

- weights are too large (leading to exploding activations or gradients),
- weights are too small (leading to vanishing activations or gradients),
- biases are improperly set (shifting activations excessively),
- parameter distributions are unbalanced across layers.

Appropriate initialization keeps variance stable across forward and backward passes, preserving meaningful gradients.

4.6.2 Key Initialization Principles

Zero Initialization (Incorrect). Setting all weights to zero forces gradients to be identical across neurons, making them learn the same function. Only biases can be initialized to zero; weights must break symmetry.

Random Initialization. Weights are sampled from predefined distributions (uniform or normal), scaled to ensure proper variance propagation. Biases are typically initialized to zero or small constants.

4.6.3 Xavier/Glorot Initialization

Designed for networks with symmetric activation functions such as `tanh` or `sigmoid`. Xavier initialization maintains variance across layers by scaling weights based on input-output dimensionality:

$$W \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right),$$

or with a normal distribution:

$$W \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}} + n_{\text{out}}}\right).$$

Purpose. Ensures that activations and gradients maintain roughly the same variance between layers, preventing early saturation of nonlinearities.

4.6.4 Kaiming/He Initialization

Designed for ReLU-based networks, which discard negative activations. The variance is adjusted accordingly:

$$W \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}}}\right) \quad \text{or} \quad W \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}}}}, \sqrt{\frac{6}{n_{\text{in}}}}\right).$$

Why it works. ReLU activation halves the effective variance by zeroing negative values; He initialization compensates for this, preserving stable forward/backward propagation.

4.6.5 Orthogonal Initialization

Orthogonal initialization creates a weight matrix W whose rows (or columns) form an orthonormal basis:

$$W^\top W = I.$$

Benefits.

- Preserves vector norms, improving stability.
- Effective in recurrent networks to counter vanishing/exploding gradients.
- Often used in architectures requiring long-range dependencies (e.g., RNNs, LSTMs).

Orthogonal initialization is especially beneficial when layer dimensions allow square or near-square matrices.

4.6.6 Initialization for Convolutional Layers

For convolutional kernels, n_{in} and n_{out} correspond to:

- $n_{\text{in}} = \text{kernel size} \times \text{input channels}$,
- $n_{\text{out}} = \text{kernel size} \times \text{output channels}$.

Thus, Xavier and He initializations extend naturally to CNNs, ensuring consistent variance across spatial dimensions.

4.6.7 Initialization in Transformers and Large Models

Transformer architectures use initialization schemes tailored to stability during deep stacking:

- Linear layers often use Xavier uniform or normal initialization.
- Residual branches may scale initial weights by $1/\sqrt{d}$ to stabilize deep residual paths.
- LayerNorm ensures controlled variance throughout the network.
- Large models sometimes use *parameter rescaling* (e.g., GPT-style $1/\sqrt{n_{\text{layers}}}$) to avoid accumulation of residual outputs.

These techniques ensure consistent gradient flow through dozens or hundreds of layers.

4.6.8 Parameterization Techniques

Weight Tying. Reusing the same parameters across multiple layers or components—for example, tying input and output embeddings in language models—reduces parameter count and improves generalization.

Spectral Normalization. Constrains the Lipschitz constant of layers by normalizing the spectral norm of weight matrices. Common in GANs to stabilize training.

Normalization and Centering. Normalization layers (BatchNorm, LayerNorm) indirectly influence parameterization by normalizing internal activations, allowing weights to focus on learning meaningful transformations rather than scale adjustments.

4.6.9 Summary

Weight initialization and parameterization determine how well signals propagate through a network during early training. Techniques such as Xavier and He initialization ensure variance stability, orthogonal matrices preserve gradient norms, and architecture-specific strategies (e.g., residual scaling in Transformers) enable very deep models to train reliably. Together, proper initialization and thoughtful parameterization form a foundation for stable, efficient, and scalable neural network optimization.

4.7 Training and Optimization

Training and optimization refer to the set of techniques, algorithms, and practices used to adjust the parameters of a neural network so that it minimizes a loss function and generalizes well to unseen data. Modern deep learning models often contain millions—or even billions—of parameters, making efficient optimization essential both for convergence and for practical deployment on resource-constrained hardware such as embedded systems and RISC-V processors.

Optimization is not merely about achieving higher accuracy: it is also about improving efficiency, reducing computational cost, and enabling models to run on low-power devices with strict memory and energy limitations.

4.7.1 Why Model Optimization Matters

Models that are not optimized suffer from:

- longer training times,
- slower inference,
- excessive memory consumption,
- reduced portability across hardware platforms,
- increased energy usage,
- poor scalability to larger datasets or architectures.

In real-world deployment scenarios—especially on embedded systems—optimization is crucial. For example, a RISC-V microcontroller might have:

- only a few hundred kilobytes of RAM,
- limited flash storage,
- no hardware acceleration,
- strict energy budgets (e.g., battery-powered or energy-harvesting devices).

Deploying a standard neural network without optimization would be infeasible under these constraints. As a result, model optimization becomes an essential step in making machine learning practical beyond high-power GPU environments.

4.7.2 Core Components of Training

Training a neural network involves iterative updates to its parameters to minimize a loss function. At the highest level, the training loop consists of:

1. Forward pass: compute predictions and loss.
2. Backward pass: compute gradients using backpropagation.

3. Parameter update: adjust weights using an optimizer (SGD, Adam, etc.).

The effectiveness of each stage depends heavily on initialization, normalization, learning rate scheduling, and architectural design (e.g., residual connections).

4.7.3 Removing Unused or Non-Connected Layers

A practical but often overlooked optimization technique involves removing non-connected, inactive, or redundant layers from a network. During experimentation or auto-ML pipelines, models may accumulate:

- layers that are never used during the forward pass,
- dead branches in multi-path architectures,
- isolated modules left from design iterations,
- residual blocks that collapse to identity mappings.

These “dangling” components waste memory, slow down training, and increase inference cost without contributing to the model’s output.

Why remove them?

- **Memory efficiency:** essential when deploying on embedded systems with limited RAM (e.g., 256 KB on many RISC-V MCUs).
- **Faster inference:** fewer operations lead to reduced latency.
- **Energy savings:** crucial in battery-powered or ultra-low-power devices.
- **Model clarity:** simplified graph structures are easier to debug and analyze.

Pruning non-connected layers is often a first step in a larger optimization pipeline.

4.7.4 Training Stability and Convergence

Several techniques ensure that the optimization process progresses smoothly:

- **Learning rate scheduling:** avoids overshooting or stagnation.
- **Gradient clipping:** prevents exploding gradients.
- **Normalization layers:** stabilize activations.
- **Weight decay:** encourages simpler solutions.
- **Warm-up phases:** particularly for Transformers.

These mechanisms ensure that gradients remain in a range that supports efficient learning.

4.7.5 Model Compression and Efficiency Techniques

Optimization extends beyond training, especially when deploying on low-power chips. Common strategies include:

Pruning. Removing weights or entire neurons/channels with little impact on output:

$$W_{ij} = 0 \quad \text{if} \quad |W_{ij}| < \tau.$$

Pruning reduces memory, compute cost, and energy consumption.

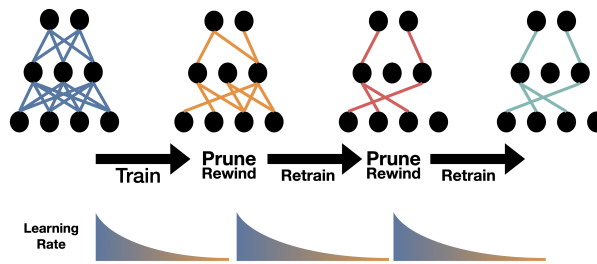


Figure 4.13: Pruning

Quantization. Reducing precision (e.g., from FP32 to INT8 or even INT4):

- significantly cuts memory usage,
- accelerates inference on simple hardware,
- reduces power consumption.

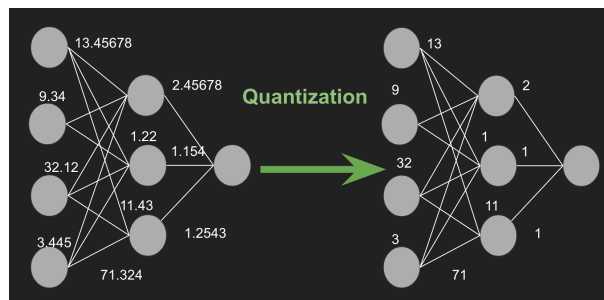


Figure 4.14: Quantization

On a RISC-V MCU, quantized integer arithmetic can outperform floating point by an order of magnitude.

Knowledge Distillation. A smaller model (student) learns to mimic a larger one (teacher), providing:

- high accuracy,
- reduced size,
- faster inference.

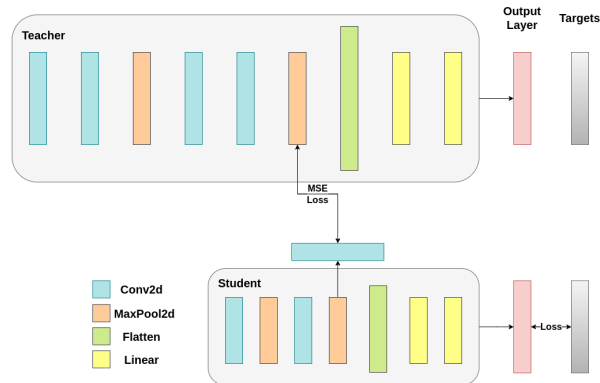


Figure 4.15: Knowledge Distillation

Operator Fusion. Combining multiple operations into one (e.g., Linear + ReLU), reducing overhead and memory access.

4.7.6 Example: Optimizing a Model for RISC-V Deployment

Deploying a CNN or MLP on a RISC-V microcontroller with 256 KB of RAM and no floating-point unit illustrates the need for optimization:

- **Pruning** removes unnecessary neurons, halving the computational load.
- **INT8 quantization** reduces model size by 4× and speeds up arithmetic.
- **Removing unused layers** prevents wasting memory on dead branches.
- **Fusion of operations** minimizes RAM usage by reducing intermediate tensors.

Without these steps, even small models would exceed memory limits or consume excessive energy—making deployment impossible.

4.7.7 Summary

Training and optimization encompass all techniques that enable neural networks to learn effectively and run efficiently. Beyond algorithmic improvements such as learning rate scheduling and gradient-based optimization, practical deployment often requires aggressive reduction of model size, removal of unused components, and adaptation to hardware constraints. In environments like RISC-V microcontrollers, these optimization strategies are not optional—they are essential for enabling machine learning applications under strict memory, latency, and energy budgets.

4.8 Loss Functions (Cross-Entropy, MSE, etc.)

Loss functions measure how well a model's predictions match the ground-truth data. They provide the training signal used by optimization algorithms to update parameters through gradient descent. Selecting an appropriate loss function is essential for stable learning, accurate predictions, and proper handling of different tasks such as classification, regression, segmentation, or ranking.

Different loss functions encode different assumptions about the data distribution and influence gradient magnitude, convergence speed, and model behavior. This section provides an overview of the most commonly used losses in deep learning.

4.8.1 Mean Squared Error (MSE)

The *Mean Squared Error* is widely used for regression tasks. Given predictions $\hat{y}$ and true values y , the MSE is defined as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2.$$

○ MEAN SQUARED ERROR

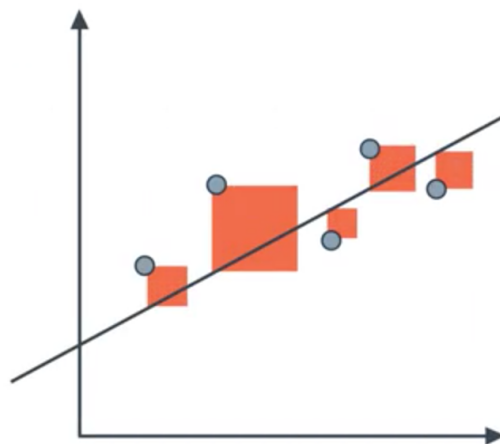


Figure 4.16: Mean Squared Error (MSE)

Intuition. MSE penalizes large errors more heavily than small ones due to the squared term. It corresponds to assuming that the data is corrupted by Gaussian noise.

Advantages.

- Simple and differentiable.
- Smooth gradient landscape.
- Strong performance when noise is Gaussian.

Limitations.

- Sensitive to outliers.
- May lead to slow convergence for very large errors.

MSE is commonly used in regression networks, autoencoders, and some generative models.

4.8.2 Mean Absolute Error (MAE)

The *Mean Absolute Error* is defined as:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|.$$

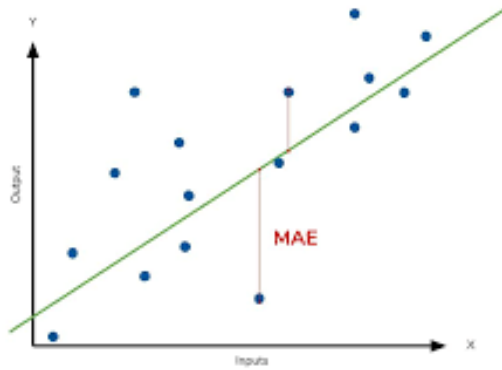


Figure 4.17: Mean Absolute Error (MAE)

Properties.

- Less sensitive to outliers than MSE.
- Produces gradients of constant magnitude.

MAE corresponds to assuming Laplacian noise and is useful in datasets with heavy-tailed distributions.

4.8.3 Huber Loss

Huber loss combines the benefits of MSE and MAE:

$$\mathcal{L}_{\delta} = \begin{cases} \frac{1}{2}(\hat{y} - y)^2 & |\hat{y} - y| \leq \delta, \\ \delta |\hat{y} - y| - \frac{1}{2}\delta^2 & |\hat{y} - y| > \delta. \end{cases}$$

It is quadratic for small errors and linear for large errors, stabilizing gradients without being overly sensitive to outliers.

Used in robotics, reinforcement learning, and safety-critical regression tasks.

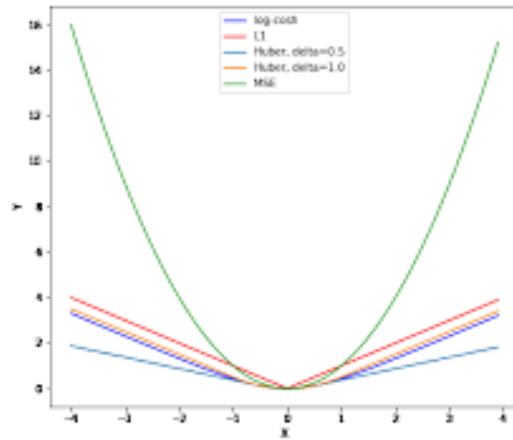


Figure 4.18: Huber Loss

4.8.4 Cross-Entropy Loss

Cross-entropy is the standard loss for classification tasks. Given a predicted probability distribution $\hat{p}$ and true distribution p , the cross-entropy is:

$$\mathcal{L}_{\text{CE}} = - \sum_{k=1}^K p_k \log(\hat{p}_k).$$

For one-hot ground truth:

$$\mathcal{L}_{\text{CE}} = -\log(\hat{p}_y),$$

where y is the correct class.

Why it works.

- Encourages correct class probability to increase.
- Produces strong gradients even when prediction confidence is low.
- Aligns with maximum likelihood estimation under categorical distribution.

Cross-entropy variants include:

- Binary Cross-Entropy (BCE) for two-class problems,
- Categorical Cross-Entropy for multi-class softmax outputs,
- BCEWithLogits for numerically stable sigmoid-based BCE.

4.8.5 Kullback–Leibler Divergence (KL Divergence)

KL divergence measures dissimilarity between two probability distributions:

$$D_{\text{KL}}(p \parallel q) = \sum_k p_k \log \left(\frac{p_k}{q_k} \right).$$

Used in:

- Variational Autoencoders (VAEs),
- Knowledge distillation,
- Probabilistic modeling.

KL divergence penalizes incorrect distributions rather than incorrect labels, making it ideal for soft-label training.

4.8.6 Contrastive and Triplet Losses

Used in metric learning to enforce similarity relationships.

Contrastive loss:

$$\mathcal{L} = y D(x_1, x_2)^2 + (1 - y) \max(0, m - D(x_1, x_2))^2.$$

Triplet loss:

$$\mathcal{L} = \max(0, D(a, p) - D(a, n) + m).$$

These losses train models to map similar inputs near each other and dissimilar ones far apart in embedding space (e.g., face recognition, retrieval systems).

4.8.7 Dice Loss and IoU Loss

Used primarily in segmentation tasks where pixel-level overlap is critical.

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2|A \cap B|}{|A| + |B|}.$$

These losses directly optimize for overlap accuracy, often outperforming cross-entropy in unbalanced segmentation.

4.8.8 Why Choosing the Right Loss Matters

Different losses affect:

- gradient structure,
- optimization speed,
- robustness to noise and outliers,

- final performance,
- probabilistic interpretation,
- architectural design (e.g., activation choice in last layer).

For example:

- Using MSE instead of cross-entropy for classification causes slow and unstable learning.
- Using cross-entropy for regression is mathematically inappropriate.
- Choosing BCE vs. BCEWithLogits determines numerical stability.

The loss should match both the task and the model’s output distribution.

4.8.9 Summary

Loss functions provide the optimization signal that drives learning. MSE, MAE, and Huber loss address regression; cross-entropy handles classification; KL divergence powers probabilistic and generative models; and metric-learning losses structure embedding spaces. Selecting the correct loss is essential for stable training, proper convergence, and ultimately achieving high predictive performance.

4.9 Optimization Algorithms (SGD, Adam, RMSProp)

Optimization algorithms govern how neural network parameters are updated to minimize a loss function. They determine the direction and magnitude of each update and thus directly affect convergence speed, stability, and final generalization. Below, we present a concise yet rigorous overview of Stochastic Gradient Descent (SGD), RMSProp, and Adam—explaining their mechanics, intuition, and when each is preferable.

4.9.1 From Gradient Descent to Stochastic Updates

Given parameters θ and loss $\mathcal{L}(\theta)$, (*full-batch*) gradient descent updates:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t),$$

where η is the learning rate. In deep learning, we approximate the gradient using mini-batches B_t :

$$g_t = \nabla_{\theta} \mathcal{L}_{B_t}(\theta_t), \quad \theta_{t+1} = \theta_t - \eta g_t,$$

which yields *Stochastic Gradient Descent (SGD)*—noisier but vastly more efficient on large datasets.

4.9.2 SGD (with Momentum and Nesterov)

Vanilla SGD. SGD performs small steps along the negative gradient. It is simple, memory-light, and often yields strong generalization, but raw SGD can be slow in narrow valleys and sensitive to learning-rate choice.

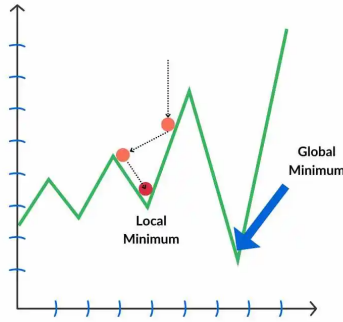


Figure 4.19: Stochastic Gradient Descent

SGD with Momentum. To accelerate along consistent directions and damp oscillations, momentum accumulates an *exponential moving average* of past gradients:

$$v_t = \mu v_{t-1} + (1 - \mu) g_t, \quad \theta_{t+1} = \theta_t - \eta v_t,$$

where v_t is the velocity and $\mu \in [0, 1)$ is the momentum coefficient (e.g., 0.9). Intuition: like pushing a heavy ball down a slope—small, consistent pushes add up to faster progress.

Nesterov Momentum (NAG). Looks ahead before computing the gradient, often yielding smoother convergence:

$$\tilde{\theta}_t = \theta_t - \eta \mu v_{t-1}, \quad v_t = \mu v_{t-1} + (1 - \mu) \nabla_{\theta} \mathcal{L}_{B_t}(\tilde{\theta}_t), \quad \theta_{t+1} = \theta_t - \eta v_t.$$

When to implement SGD.

- **Strong baseline & generalization:** In vision tasks (e.g., CNNs), SGD(+momentum) often reaches better generalization than adaptive methods.
- **Stable loss landscapes:** When gradients are relatively well-scaled (after normalization), SGD can be very effective.
- **Low memory footprint:** Minimal optimizer state.

A simple example. Minimize $f(\theta) = \frac{1}{2}a\theta^2$ with $a > 0$. SGD update: $\theta_{t+1} = \theta_t - \eta a \theta_t = (1 - \eta a)\theta_t$. If η is too large, $(1 - \eta a)$ oscillates or diverges; with momentum, updates dampen oscillations along steep directions (large a).

4.9.3 RMSProp

Idea. RMSProp adapts the learning rate per-parameter by normalizing the gradient with a running average of its squared values—mitigating issues where some parameters consistently have larger gradients.

Update rule. With decay $\beta \in (0, 1)$ (e.g., 0.9) and small ϵ (e.g., 10^{-8}):

$$s_t = \beta s_{t-1} + (1 - \beta) g_t^2, \quad \theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{s_t + \epsilon}} g_t.$$

Here s_t tracks recent gradient magnitudes; parameters with consistently large gradients get effectively smaller step sizes, and vice versa.

When to implement RMSProp.

- **Non-stationary, noisy problems:** Historically popular for RNNs and reinforcement learning where gradient scales change over time.
- **Unevenly scaled features:** Works well when different parameters exhibit very different gradient magnitudes.
- **Fast tuning:** Fewer LR sweeps needed compared to raw SGD.

Toy intuition. If one parameter sees gradients about $10\times$ larger than another, RMSProp will downscale its effective step, helping keep updates balanced without manual per-parameter tuning.

4.9.4 Adam (and AdamW)

Idea. Adam combines momentum (first-moment averaging) with RMSProp-like scaling (second-moment averaging), plus bias corrections—yielding robust, per-parameter adaptive steps.

Update rule. With (β_1, β_2) (e.g., 0.9, 0.999) and ϵ :

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2,$$

Bias-corrected estimates:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t},$$

Parameter update:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t.$$

AdamW (decoupled weight decay). Standard Adam mixes L_2 regularization into the adaptive scaling, which can behave differently from true weight decay. AdamW decouples the decay:

$$\theta_{t+1} = (1 - \eta\lambda) \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t,$$

where λ is the weight decay coefficient—often preferred in modern deep nets (e.g., Transformers).

When to implement Adam/AdamW.

- **Strong default for many tasks:** Fast convergence and ease of tuning.
- **Sparse or heavy-tailed gradients:** NLP, Transformers, embeddings—adaptive steps help a lot.
- **Heterogeneous scales:** Different layers/parameters with widely varying gradient magnitudes.

Caveats.

- **Generalization gap:** On some vision tasks, Adam may converge faster but generalize slightly worse than tuned SGD(+momentum).
- **Tune ϵ if needed:** Can stabilize training when variances are tiny.
- **Prefer AdamW:** Decoupled weight decay tends to yield more predictable regularization.

4.9.5 Choosing Between SGD, RMSProp, and Adam

- **Use SGD (+Momentum/Nesterov)** when you care about *final generalization* and have reasonably normalized inputs (common in CNN-based vision). It is memory-efficient and often wins with enough tuning (LR, momentum, schedule).
- **Use RMSProp** for non-stationary or very noisy objectives (e.g., some RL or RNN settings) and when gradient scales drift significantly over training.
- **Use Adam/AdamW** as a strong, practical default—particularly in NLP/Transformers, multimodal models, and problems with sparse or uneven gradients—fast to good performance with modest tuning.

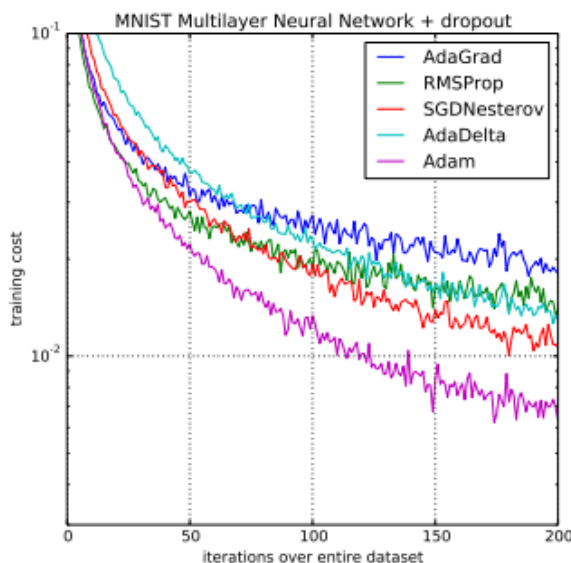


Figure 4.20: Comparison between AdaGrad, RMSProp, SGD Nesterov, AdaDelta and Adam

4.9.6 Hyperparameter Heuristics and Practical Tips

- **Defaults:** Adam/AdamW: $\eta=10^{-3}$, $(\beta_1, \beta_2)=(0.9, 0.999)$, $\epsilon=10^{-8}$; RMSProp: $\eta=10^{-3}$, $\beta=0.9$, $\epsilon=10^{-8}$; SGD: start $\eta \in [10^{-2}, 10^{-1}]$, momentum 0.9.

- **Schedule the LR:** Step decay or cosine annealing often improves all three optimizers.
- **Normalize/standardize inputs:** Helps SGD; still beneficial for adaptive methods.
- **Batch size coupling:** Larger batches often need larger learning rates (linear scaling rule).
- **Regularization:** Prefer decoupled weight decay (AdamW) over L_2 in Adam; with SGD use weight decay and/or dropout.

4.9.7 Intuition via a Simple 2D Bowl

Consider minimizing $f(\theta_1, \theta_2) = \frac{1}{2}(a\theta_1^2 + b\theta_2^2)$ with $a \gg b$ (a narrow, steep valley).

- **SGD** zig-zags across the steep axis unless momentum is used; with momentum it accelerates along the shallow axis and damps oscillations along the steep one.
- **RMSProp** scales steps by recent squared gradients, shrinking steps along the steep axis (large gradients) and enlarging along the shallow axis—quickly balancing progress.
- **Adam** couples momentum (directional memory) with RMSProp-like scaling, often moving decisively toward the minimum with stable steps; AdamW improves regularization on top.

4.9.8 Summary

SGD, RMSProp, and Adam represent a progression from simple, global step sizes to adaptive, per-parameter updates with momentum. **SGD(+momentum)** remains a gold standard for strong generalization (especially in vision). **RMSProp** addresses non-stationary, uneven gradient scales. **Adam/AdamW** is a robust default for many modern architectures, particularly where gradients are sparse or heterogeneous. Selecting among them—and combining with a sound learning-rate schedule—often determines whether training is merely adequate or state-of-the-art.

4.10 Learning Rate Scheduling

The *learning rate* is one of the most fundamental hyperparameters in the optimization of neural networks. It determines the size of the steps taken by the optimizer when updating the model’s parameters in response to the computed gradients. In simple terms, the learning rate controls how quickly or cautiously a model learns from the training data.

At each iteration of gradient-based optimization, the model parameters θ are updated according to:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta_t} \mathcal{L}(\theta_t),$$

where $\mathcal{L}(\theta_t)$ is the loss function, $\nabla_{\theta_t}\mathcal{L}$ is the gradient of the loss with respect to the parameters, and η is the learning rate. A larger η means taking bigger steps in the direction that reduces the loss, while a smaller η means taking smaller, more cautious steps.

The learning rate directly affects the trajectory of the optimization process through the loss landscape:

- If the learning rate is **too high**, the optimization steps overshoot the minima, leading to oscillations or even divergence of the loss.
- If it is **too low**, progress becomes extremely slow, and the model might get trapped in shallow minima or plateaus.

Thus, selecting an appropriate learning rate is a critical factor in achieving efficient and stable convergence. However, a single static value rarely performs optimally throughout training—different phases of learning benefit from different update magnitudes. This observation motivates the use of *learning rate scheduling*.

4.10.1 Theoretical Overview of Learning Rate Scheduling

Learning rate scheduling refers to the process of dynamically adjusting the learning rate η_t as training progresses. Early in training, a relatively large learning rate helps the model explore the loss surface and escape poor local minima. Later in training, a smaller learning rate allows for more precise fine-tuning as the optimizer approaches convergence.

Mathematically, the update rule with a time-dependent learning rate becomes:

$$\theta_{t+1} = \theta_t - \eta_t \nabla_{\theta_t} \mathcal{L}(\theta_t),$$

where η_t evolves according to a scheduling policy. Different scheduling strategies define how η_t changes over epochs or iterations.

Common scheduling methods include:

- **Step Decay:** Reduces the learning rate by a constant factor after a fixed number of epochs:

$$\eta_t = \eta_0 \cdot \gamma^{\lfloor t/T \rfloor}.$$

- **Exponential Decay:** Decreases the learning rate continuously as $\eta_t = \eta_0 \cdot e^{-\lambda t}$, providing a smooth decay.
- **Cosine Annealing:** Gradually lowers the learning rate following a cosine curve:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T_{\max}} \pi \right) \right).$$

- **Cyclical Learning Rate (CLR):** Periodically increases and decreases η_t between lower and upper bounds to escape local minima.
- **One-Cycle Policy:** Increases the learning rate rapidly at the start of training, then gradually reduces it below the initial value, promoting fast convergence and strong generalization.

A *warm-up* phase is often included, during which the learning rate starts small and increases linearly or exponentially for the first few epochs, stabilizing early training—especially in deep architectures such as Transformers.

4.10.2 Influence of the Learning Rate on Model Training

The learning rate profoundly influences how a neural network learns. Its value and evolution determine whether training will be stable, efficient, or divergent.

- **High Learning Rate:** When η is too large, parameter updates overshoot the optimal region, causing oscillations or divergence. The model may fail to minimize the loss and instead fluctuate around high-error regions. While a large learning rate accelerates initial progress, it often prevents fine convergence.
- **Low Learning Rate:** If η is too small, updates become negligible, and training proceeds very slowly. The model may converge prematurely to suboptimal minima or fail to adapt effectively, leading to underfitting.
- **Dynamic Scheduling:** A learning rate schedule allows a controlled transition—from fast exploration to fine-grained optimization. Larger learning rates early on encourage broad exploration of the loss surface, while progressively smaller rates near convergence allow the model to refine weights precisely. This results in both faster training and improved generalization.

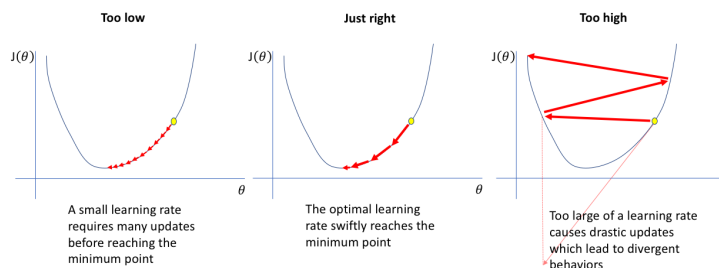


Figure 4.21: Learning Rate explained

The relationship between learning rate and other hyperparameters—such as batch size, momentum, and weight decay—is also crucial. For instance, larger batch sizes often require proportionally higher learning rates to maintain gradient stability, a phenomenon described by the *linear scaling rule*.

4.10.3 Practical Considerations

In practice, identifying an optimal learning rate or schedule typically involves empirical experimentation. A common approach is the *learning rate finder*, which gradually increases the learning rate during a brief training run and records the loss. Plotting loss against learning rate reveals a region where the loss decreases most rapidly—indicating an effective initial value.

Deep learning frameworks such as PyTorch and TensorFlow provide built-in tools for scheduling, including: `StepLR`, `ExponentialLR`, `ReduceLROnPlateau`, and `CosineAnnealingLR`. These utilities automate the adjustment of η_t during training, making learning rate scheduling a standard part of modern optimization pipelines.

In summary, the learning rate defines the pace of learning in neural network training—too high leads to instability, too low to stagnation. Learning rate scheduling refines this process by allowing the model to adaptively balance exploration and convergence, resulting in faster, more stable, and more accurate learning.

4.11 Early Stopping and Regularization During Training

Training deep neural networks is an iterative optimization process that requires a delicate balance between learning the underlying patterns of the data and memorizing its noise. While optimization algorithms aim to minimize the training loss indefinitely, the ultimate goal of machine learning is generalization to unseen data.

Early Stopping is a regularization technique that addresses this discrepancy by monitoring the model's performance on a validation set and terminating training when generalization performance begins to degrade. Unlike structural regularization methods (like Dropout) or penalty terms (like Weight Decay), early stopping operates purely at the procedural level of the training loop.

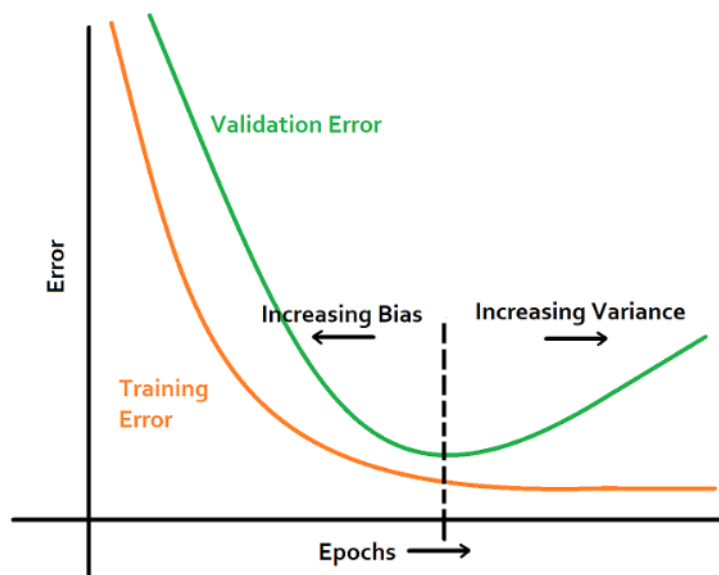


Figure 4.22: Early Stopping Concept

4.11.1 The Dynamics of Overfitting

As training progresses, a neural network typically goes through two distinct phases:

1. **Learning Phase:** Initially, both training loss and validation loss decrease. The model learns general features (e.g., edges, shapes, syntax) that are applicable to both the training and validation sets. The parameters move from their initial random state towards a region of low error.

2. **Overfitting Phase:** After a certain number of epochs, the training loss continues to decrease, often approaching zero. However, the validation loss reaches a minimum (the inflection point) and starts to rise. At this stage, the model begins to co-adapt to the specific noise or idiosyncrasies of the training data, losing its ability to generalize.

Early stopping aims to identify this specific inflection point t^* where the validation risk is minimized:

$$t^* = \operatorname{argmin}_t \mathcal{L}_{val}(\theta_t).$$

4.11.2 Implementation Strategy: Patience and Delta

Because optimization with Stochastic Gradient Descent (SGD) is noisy, the validation loss rarely follows a smooth, monotonic curve. It often fluctuates due to batch sampling variance. Stopping immediately at the first sign of increased loss would effectively halt training prematurely.

To handle this stochasticity, robust implementations introduce two key hyperparameters:

- **Patience (p):** This defines a "tolerance window." It is the number of epochs the training is allowed to continue without observing an improvement in the validation metric. For example, with $p = 10$, the model has 10 epochs to beat the previous best score before training is terminated. This allows the optimizer to potentially escape local plateaus.
- **Minimum Delta (δ):** This threshold defines what counts as a "significant" improvement. A decrease in loss smaller than δ is considered noise and does not reset the patience counter.

Checkpointing (Best vs. Last Model): Crucially, when early stopping triggers, the model's parameters at the final epoch (θ_{final}) are often suboptimal (since the loss has likely risen). Therefore, a proper implementation must perform *checkpointing*: saving the weights θ_{best} every time the validation metric improves, and reloading these weights when training stops.

4.11.3 Early Stopping as Implicit Regularization

Theoretical analysis suggests that early stopping acts as a regularizer similar to L_2 Weight Decay. When weights are initialized with small values (centered around zero), they grow in magnitude as training proceeds to minimize the loss. By stopping training at time t , we effectively restrict the optimization to a volume of the parameter space reachable within t steps.

This restriction prevents the weights from growing excessively large, which is often a symptom of overfitting (where large weights create sharp decision boundaries to fit outliers). Thus, the number of training epochs effectively becomes a hyperparameter that controls model complexity.

4.11.4 Interaction with Other Techniques

Early stopping is rarely used in isolation. It functions best when combined with:

- **Learning Rate Scheduling:** Often, when validation loss plateaus, reducing the learning rate (e.g., `ReduceLROnPlateau`) can yield further improvements before early stopping is necessary.
- **Dropout and Weight Decay:** These techniques slow down the onset of overfitting, allowing the model to train for more epochs and learn more complex features before early stopping is triggered.

4.11.5 Summary

Early stopping provides a computationally efficient and intuitive safeguard against overfitting. By utilizing a separate validation set to monitor generalization error, it allows practitioners to train models without knowing the optimal number of epochs in advance. It ensures that the final model represents the best balance between underfitting and overfitting, retrieving the optimal parameter configuration θ_{best} rather than the final, potentially overfitted state.

4.12 Gradient Clipping and Stability Techniques

Training deep neural networks, particularly Recurrent Neural Networks (RNNs) or very deep Transformers, can be destabilized by the **exploding gradient problem**. This phenomenon occurs when large error gradients accumulate during backpropagation, resulting in very large updates to the model weights.

These massive updates can cause the numerical values of the parameters to overflow (NaNs) or shoot into sub-optimal regions of the loss landscape from which the model cannot recover. **Gradient Clipping** is the standard technique used to counteract this instability by limiting the magnitude of the gradients before the optimization step.

4.12.1 The Exploding Gradient Problem

In deep architectures, the gradient is computed via the chain rule, involving the repeated multiplication of matrices (in feedforward nets) or the same matrix (in RNNs). If the spectral norm (largest singular value) of the weight matrices is greater than 1, the gradients can grow exponentially with the depth of the network (or the length of the sequence).

Mathematically, if the gradient vector g grows uncontrollably ($\|g\| \rightarrow \infty$), the parameter update $\Delta\theta = -\eta \cdot g$ becomes too large, destroying the learned representations.

4.12.2 Gradient Clipping Strategies

There are two primary methods to clip gradients: clipping by value and clipping by norm.

Clipping by Value (Element-wise)

This method clips each element of the gradient vector independently to a predefined range $[-\delta, \delta]$. Given a gradient vector $g \in \mathbb{R}^d$:

$$g_i \leftarrow \min(\max(g_i, -\delta), \delta)$$

for each element i .

Pros: Simple to implement.

Cons: It changes the direction of the gradient vector. If one component is clipped and another is not, the resulting vector points in a slightly different direction than the true steepest descent, potentially slowing down convergence.

Clipping by Norm (Global Norm)

This is the preferred method in modern deep learning (e.g., used in Transformers and LSTMs). It rescales the entire gradient vector g so that its L_2 norm does not exceed a threshold C .

The operation is defined as:

$$g \leftarrow g \cdot \frac{C}{\max(C, \|g\|_2)}$$

where $\|g\|_2$ is the Euclidean norm of the vector.

Mechanism:

- If $\|g\|_2 \leq C$, the scaling factor is 1, and the gradient is unchanged.
- If $\|g\|_2 > C$, the gradient is scaled down to have a norm of exactly C .

Pros: Preserves Direction. Unlike value clipping, this rescaling operation ensures that the gradient points in the exact same direction as the original error signal, only its magnitude is reduced. This is crucial for optimization stability.

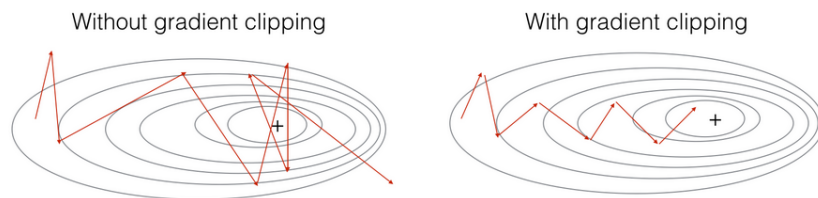


Figure 4.23: Visual comparison: Clipping by Value (changes direction) vs. Clipping by Norm (preserves direction).

4.12.3 Practical Implementation and Use Cases

When to use:

- **RNNs / LSTMs:** Essential. Without clipping, training on long sequences (Back-propagation Through Time) often fails.
- **Transformers:** Standard practice (e.g., global norm clip = 1.0) to stabilize the pre-training of large language models.
- **High Learning Rates:** Clipping allows the use of larger learning rates by putting a safety cap on the worst-case update size.

Choosing the threshold C : A common heuristic is to look at the average gradient norm over an epoch and set C to a value slightly higher (e.g., 5.0 or 1.0). If C is too small, it may suppress useful signals; if too large, it fails to prevent explosions.

4.12.4 Summary

Gradient Clipping acts as a "safety valve" for the optimization process. By enforcing a maximum limit on the gradient magnitudes—preferably via Norm Clipping to preserve direction—it ensures that parameter updates remain within a reasonable range. This technique effectively solves the exploding gradient problem, enabling the successful training of deep and recurrent architectures that would otherwise be numerically unstable.

4.13 Transfer Learning and Fine-Tuning

Transfer Learning is a methodology that leverages the knowledge gained from solving one problem and applies it to a different but related problem. Instead of training a deep neural network from scratch (tabula rasa)—which requires massive datasets and significant computational resources—transfer learning initializes the network with weights learned from a large-scale source task (e.g., ImageNet classification).

This approach relies on the hierarchical nature of deep learning: early layers learn generic, low-level features (edges, textures, colors), while deeper layers learn high-level, task-specific semantic representations.

4.13.1 Core Paradigms

There are two primary strategies for adapting a pre-trained model to a target task:

Feature Extraction (Freezing)

In this approach, the pre-trained convolutional base (or encoder) is treated as a fixed feature extractor.

- **Mechanism:** The weights of the backbone layers are "frozen" (i.e., excluded from gradient updates). Only the final classification head (fully connected layers) is initialized randomly and trained on the target dataset.
- **Use Case:** Ideal when the target dataset is small and similar to the source dataset, or when computational resources are strictly limited.

Fine-Tuning

Fine-tuning involves unfreezing some or all of the layers of the pre-trained base and jointly training them with the new classification head.

- **Mechanism:** Gradients are backpropagated through the entire network (or specific blocks). This allows the pre-trained features to adapt slightly to the nuances of the new data.
- **Use Case:** Preferred when the target dataset is large, or when the domain differs significantly from the source, requiring the adaptation of deeper representations.

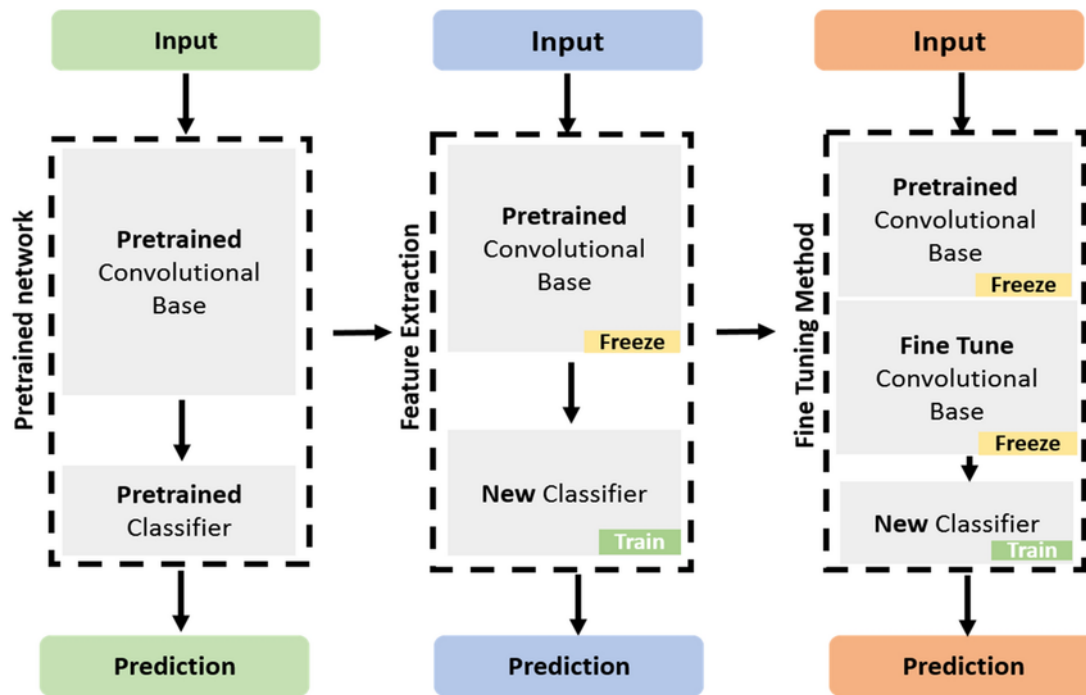


Figure 4.24: Comparison: Feature Extraction (Frozen Backbone) vs. Fine-Tuning (Trainable Backbone).

4.13.2 Strategic Guidelines

The choice between freezing and fine-tuning depends on two factors: the size of the target dataset and its similarity to the source dataset.

1. **Small Data, Similar Domain:** *Strategy: Freeze Backbone.* Fine-tuning runs a high risk of overfitting. The pre-trained features are already relevant; training a linear classifier on top is sufficient.
2. **Large Data, Similar Domain:** *Strategy: Fine-Tune.* With sufficient data, overfitting is less of a concern. Unfreezing the whole network allows the model to maximize performance by specializing features.
3. **Small Data, Different Domain:** *Strategy: Freeze Early Layers, Train Later Layers.* Early layers detect generic features (edges) useful everywhere. Later layers detect specific objects (e.g., dog breeds) not useful for the target (e.g., medical cells). Train only the later blocks.
4. **Large Data, Different Domain:** *Strategy: Train from Scratch or Full Fine-Tuning.* Since the domains are distinct, pre-trained high-level features may be misleading. However, initializing with pre-trained weights usually converges faster than random initialization.

4.13.3 Implementation Techniques

Differential Learning Rates (Discriminative Fine-Tuning)

When fine-tuning, it is often suboptimal to use the same learning rate for all layers. The early layers contain robust, general knowledge that should be preserved, while the new

head needs to learn rapidly.

- **Technique:** Apply a lower learning rate η_{base} to the backbone (e.g., 10^{-5}) and a higher learning rate η_{head} to the classifier (e.g., 10^{-3}).
- **Formula:** $\eta^l = \eta_{base} \cdot \xi^l$, where ξ is a decay factor across layers l .

Gradual Unfreezing

To prevent "catastrophic forgetting"—where massive gradient updates from the randomly initialized head destroy the pre-trained weights in the backbone—layers can be unfrozen sequentially.

1. Train only the head for a few epochs.
2. Unfreeze the last block of the backbone and train with a reduced learning rate.
3. Gradually unfreeze earlier blocks until the whole network is trainable.

4.13.4 Summary

Transfer learning is the catalyst that democratized deep learning, allowing state-of-the-art performance on specific tasks with limited data. By understanding the trade-off between freezing (preserving generic knowledge) and fine-tuning (specializing for the target), practitioners can drastically reduce training time and improve generalization.

4.14 Implementation Patterns and Practical Considerations

Translating mathematical models into working code requires more than just transcribing equations. It involves adopting robust software engineering patterns that ensure reproducibility, scalability, and ease of debugging. A well-implemented neural network code-base abstracts hardware specifics, handles data efficiently, and guards against numerical instability.

4.14.1 Reproducibility and Seeding

Deep learning frameworks rely heavily on pseudo-random number generators (PRNGs) for weight initialization, data shuffling, and dropout masks. To ensure that experiments are reproducible—meaning a re-run produces the exact same results—one must fix the seeds across all libraries.

A robust seeding pattern involves setting seeds for:

1. The core language random module (e.g., Python `random`).
2. The numerical library (e.g., `numpy`).
3. The deep learning framework (e.g., `torch.manual_seed`).
4. The hardware backend (e.g., `torch.cuda.manual_seed_all`).

Additionally, ensuring deterministic algorithms on GPUs (e.g., setting `torch.backends.cudnn.deterministic = True`) avoids nondeterministic behavior in convolution operations, though often at a slight performance cost.

4.14.2 Device-Agnostic Code Patterns

Modern deep learning code must run seamlessly across different hardware accelerators (NVIDIA GPUs, Apple MPS, TPUs, or CPUs) without requiring code changes. Hard-coding `.cuda()` is a deprecated anti-pattern.

The Recommended Pattern: Define a global device object early in the script:

```
device = torch.device(
    "cuda" if torch.cuda.is_available() else
    "mps" if torch.backends.mps.is_available() else
    "cpu"
)
```

All models and tensors should be moved to this device using the `.to(device)` method. This ensures portability and simplifies deployment on diverse infrastructure.

4.14.3 Numerical Stability Patterns

Floating-point arithmetic is not infinite precision. Deep networks often deal with very small numbers (probabilities) or very large ones (unnormalized logits), leading to underflow or overflow.

Log-Sum-Exp Trick: When computing the softmax of a vector x , calculating e^{x_i} can overflow if x_i is large. The stable implementation uses the identity:

$$\log\left(\sum e^{x_i}\right) = \max(x) + \log\left(\sum e^{x_i - \max(x)}\right)$$

Subtracting the maximum value ensures all exponents are ≤ 0 , guaranteeing stability. Frameworks handle this automatically in functions like `CrossEntropyLoss` (which combines `LogSoftmax` and `NLLLoss`), but manual implementations must be careful.

Epsilon for Division: When normalizing (e.g., in BatchNorm or attention), always add a small ϵ (e.g., 10^{-8}) to the denominator to prevent division by zero:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

4.14.4 Gradient Accumulation

Training large models on hardware with limited VRAM (e.g., consumer GPUs) often prevents the use of the desired batch size. Gradient accumulation decouples the *logical batch size* from the *physical batch size*.

Mechanism: Instead of updating weights after every forward-backward pass, gradients are accumulated over N mini-batches:

1. Perform forward pass and loss computation.
2. Scale loss by $1/N$.

3. Perform backward pass (`loss.backward()`), which accumulates gradients into `.grad`.
4. Repeat N times.
5. Perform optimizer step (`optimizer.step()`) and zero gradients (`optimizer.zero_grad()`).

This simulates a batch size of $N \times \text{physical_batch_size}$ with constant memory footprint.

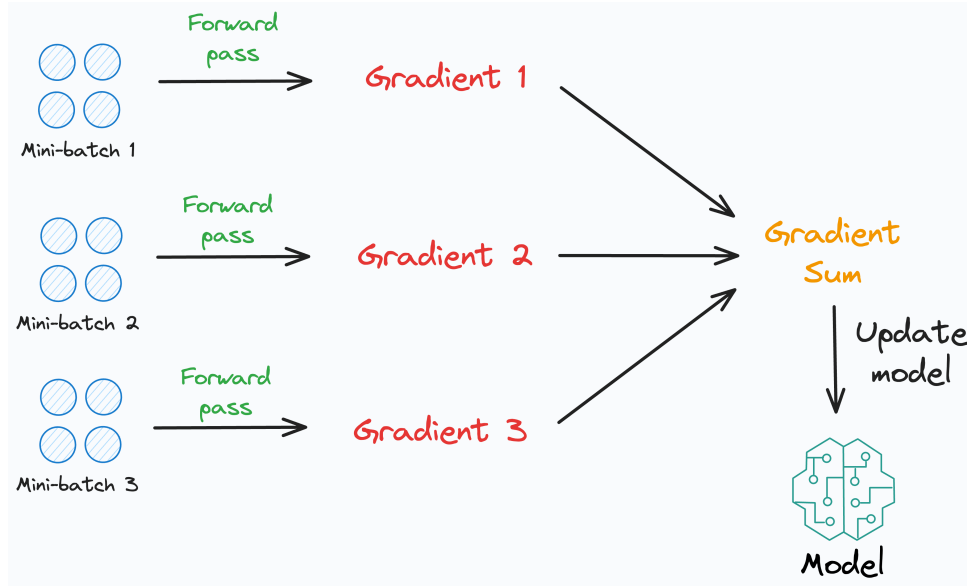


Figure 4.25: Gradient Accumulation: Aggregating gradients over multiple steps to simulate a larger batch.

4.14.5 Summary

Implementation patterns distinguish research scripts from production-grade libraries. By enforcing strict seeding for reproducibility, abstracting hardware logic, and proactively handling numerical instability, practitioners create codebases that are robust, portable, and easier to debug. These "silent" engineering decisions are often as critical to model performance as the architecture itself.

4.15 Model Definition in PyTorch (Modules, Forward Pass, Parameters)

In PyTorch, the fundamental building block for all neural network components is the `nn.Module` class. Whether defining a single layer, a complex block (like a Residual Block), or an entire architecture (like ResNet), the design pattern remains consistent: state is defined in the initialization, and computation is defined in the forward pass.

This object-oriented approach leverages PyTorch's dynamic computational graph (define-by-run), allowing for flexible architectures where control flow (loops, conditions) can be used directly within the model definition.

4.15.1 The nn.Module Anatomy

A custom model typically inherits from `torch.nn.Module` and overrides two essential methods:

- `__init__(self)`: The constructor. This is where the model's **state** is initialized. You define the layers (submodules) and parameters here. PyTorch automatically registers these attributes to track gradients and handle device movement (CPU/GPU).
- `forward(self, x)`: The computation logic. This method defines how the input tensor x flows through the initialized layers to produce an output. It is called automatically when the model instance is called (i.e., `model(x)`), effectively constructing the dynamic graph.

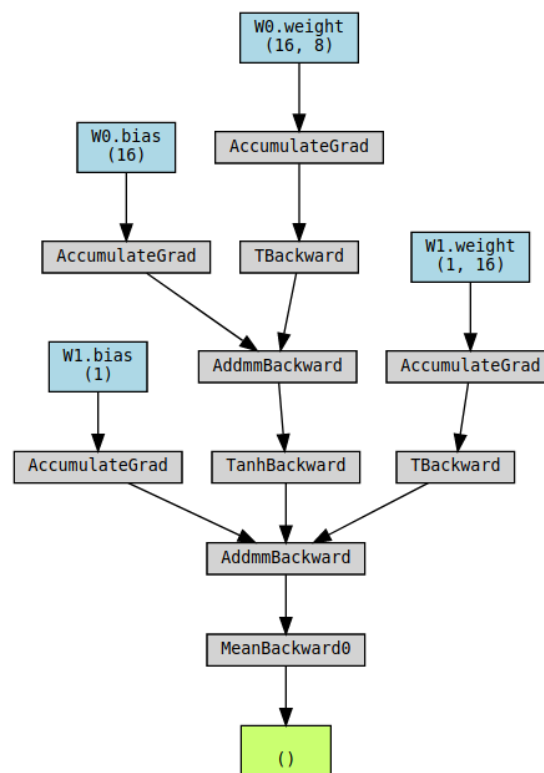


Figure 4.26: The `nn.Module` lifecycle: State registration in `__init__` and Graph construction in `forward`.

4.15.2 Parameters and Buffers

PyTorch distinguishes between two types of state variables stored within a module:

1. **Parameters (`nn.Parameter`):** Learnable tensors (weights and biases) that require gradients. They are automatically added to the list returned by `model.parameters()`, which is passed to the optimizer.

2. **Buffers (register_buffer):** Tensors that are part of the model's state but are **not** trained via gradient descent. Common examples include the running mean and variance in Batch Normalization layers or causal masks in Transformers. They persist in the `state_dict` but do not require gradients.

4.15.3 Example: A Simple MLP Block

Below is a canonical implementation of a Multilayer Perceptron (MLP) classification block.

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class SimpleMLP(nn.Module):
    def __init__(self, input_dim, hidden_dim, num_classes):
        super(SimpleMLP, self).__init__()
        # 1. Define Submodules (State)
        self.fc1 = nn.Linear(input_dim, hidden_dim)
        self.bn1 = nn.BatchNorm1d(hidden_dim)
        self.dropout = nn.Dropout(p=0.5)
        self.fc2 = nn.Linear(hidden_dim, num_classes)

        # 2. Initialize Weights (Optional explicit initialization)
        nn.init.kaiming_normal_(self.fc1.weight)

    def forward(self, x):
        # 3. Define Computation Graph
        # x shape: (Batch_Size, Input_Dim)
        x = self.fc1(x)
        x = self.bn1(x)
        x = F.relu(x) # Functional API for stateless ops
        x = self.dropout(x)
        logits = self.fc2(x)
        return logits
```

4.15.4 The Functional API vs. Module API

PyTorch provides two ways to apply operations:

- **Module API (torch.nn):** Stateful layers (e.g., `nn.Linear`, `nn.Conv2d`). These store weights and must be defined in `__init__`.
- **Functional API (torch.nn.functional):** Stateless operations (e.g., `F.relu`, `F.softmax`). These are purely mathematical functions applied directly in `forward`.

Best Practice: Use `nn.Module` for anything with learnable weights; use functional calls for simple activation functions or operations without parameters (like pooling) to keep `__init__` clean.

4.15.5 Summary

The `nn.Module` abstraction is central to PyTorch’s flexibility. By decoupling parameter storage (in `__init__`) from graph execution (in `forward`), it enables the creation of complex, dynamic architectures that are easy to debug and extend. Understanding how parameters and buffers are registered is crucial for managing model saving, loading, and optimization correctly.

4.16 Training Loops and Evaluation Pipelines

The training loop is the engine of any deep learning application. It is the iterative process where the model processes data, computes errors, and updates its parameters to minimize the loss function. While frameworks like PyTorch Lightning or Hugging Face Trainer abstract this complexity, understanding the raw loop is essential for debugging and implementing custom training logic.

A robust pipeline consists of two distinct phases executed at every epoch: a **training phase** to optimize weights, and a **validation phase** to measure generalization performance.

4.16.1 The Standard Training Loop

The training loop iterates over the dataset in batches. For each batch, the following sequence of operations must occur, typically in this exact order:

1. **Data Preparation:** Fetch a batch of inputs (x) and targets (y) from the `DataLoader` and move them to the target device (GPU/TPU).
2. **Forward Pass:** Pass inputs through the model to get predictions ($\hat{y}$).
3. **Loss Computation:** Calculate the discrepancy between $\hat{y}$ and y using the loss function.
4. **Backward Pass:** Compute gradients via backpropagation (`loss.backward()`).
5. **Parameter Update:** Adjust weights using the optimizer (`optimizer.step()`).
6. **Gradient Reset:** Clear the gradients for the next iteration (`optimizer.zero_grad()`).

4.16.2 The Evaluation Pipeline

The evaluation (or validation) pipeline is similar to training but differs in two critical aspects:

- **No Gradient Computation:** Gradients are unnecessary for inference. Wrapping code in a `torch.no_grad()` context manager disables the autograd engine, reducing memory usage and speeding up computation.
- **Model Mode (`model.eval()`):** Certain layers behave differently during training and inference. For example, **Dropout** is disabled during evaluation, and **Batch Normalization** uses running statistics instead of batch statistics. Failing to switch modes leads to incorrect metric calculation.

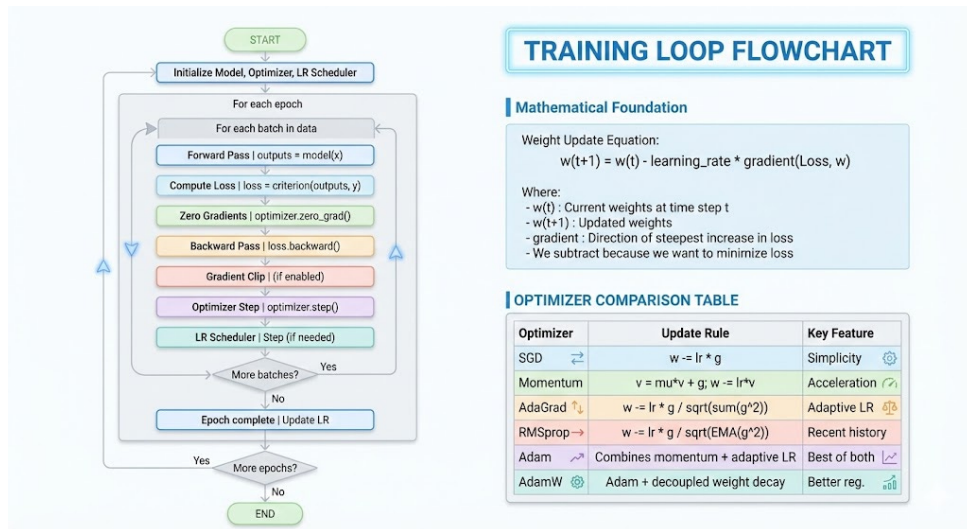


Figure 4.27: Flowchart of a standard Training Loop and Evaluation Pipeline.

4.16.3 Implementation Example

Below is a standard PyTorch implementation covering both phases.

```
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train() # 1. Set Train Mode
    running_loss = 0.0

    for inputs, targets in loader:
        inputs, targets = inputs.to(device), targets.to(device)

        # 2. Optimization Steps
        optimizer.zero_grad() # Clear gradients
        outputs = model(inputs) # Forward
        loss = criterion(outputs, targets) # Compute Loss
        loss.backward() # Backward
        optimizer.step() # Update

        running_loss += loss.item() * inputs.size(0)

    return running_loss / len(loader.dataset)

def evaluate(model, loader, criterion, device):
    model.eval() # 1. Set Eval Mode
    running_loss = 0.0
    correct = 0

    # 2. Disable Gradient Calculation
    with torch.no_grad():
        for inputs, targets in loader:
            inputs, targets = inputs.to(device), targets.to(device)
            outputs = model(inputs)
```

```

loss = criterion(outputs, targets)

running_loss += loss.item() * inputs.size(0)
preds = outputs.argmax(dim=1)
correct += (preds == targets).sum().item()

accuracy = correct / len(loader.dataset)
return running_loss / len(loader.dataset), accuracy

```

4.16.4 Best Practices

- **Gradient Zeroing Location:** Use `zero_grad(set_to_none=True)` for a slight performance boost (avoids writing zeros to memory).
- **Metric Accumulation:** Avoid storing tensors (e.g., list of predictions) on the GPU during the loop; move them to CPU or convert to numpy to prevent OOM (Out Of Memory) errors.
- **Progress Tracking:** Wrap loaders with `tqdm` for real-time progress bars and log metrics to tools like TensorBoard or Weights & Biases.

4.16.5 Summary

The training loop translates mathematical optimization into code. While conceptually simple, it requires strict discipline regarding mode switching (`.train()` vs `.eval()`) and gradient management (`no_grad()`) to ensure that the model learns correctly and is evaluated fairly on unseen data.

4.17 Saving, Loading, and Deployment of Models

4.18 Performance Profiling and Debugging

Chapter 5

Metrics & how to read it

5.1 Accuracy & Loss

During the training of a machine learning model, *accuracy* and *loss* are two key metrics used to assess how well the model is learning and generalizing. Together, they provide complementary insights: accuracy measures the proportion of correct predictions, while loss quantifies the numerical difference between the predicted and true outputs. Monitoring these metrics over time allows practitioners to diagnose issues such as underfitting, overfitting, or poor optimization behavior.

Accuracy and loss are typically visualized as curves across training epochs. The shape and trends of these curves reveal how the model's performance evolves during training and validation, helping to identify whether the model is improving, stagnating, or degrading.

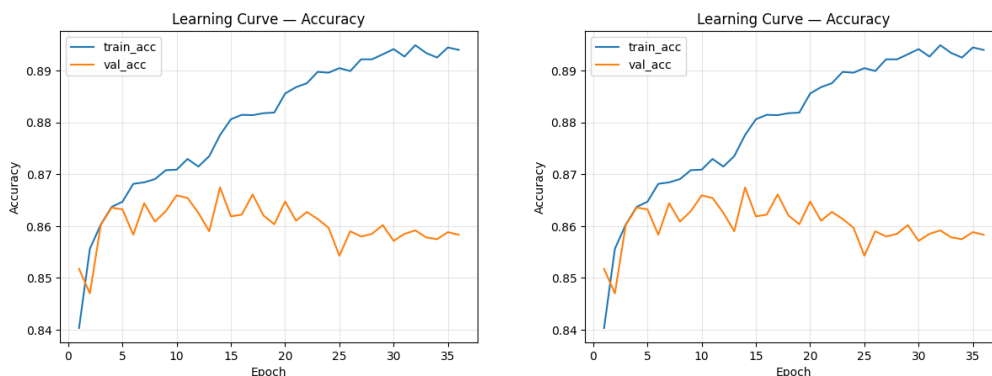


Figure 5.1: Learning Curve for Loss and Accuracy

5.1.1 Accuracy

Accuracy measures the fraction of correctly predicted samples relative to the total number of samples. For a classification task, it is formally defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}.$$

A high accuracy indicates that the model correctly classifies most examples in the dataset. However, accuracy alone can be misleading—especially in imbalanced datasets where some

classes are more frequent than others. In such cases, complementary metrics such as precision, recall, and F1-score provide a more balanced evaluation.

The *accuracy curve* plots model accuracy over successive training epochs for both training and validation sets. A steadily increasing training accuracy accompanied by stable or improving validation accuracy indicates effective learning. Conversely, if validation accuracy stagnates or declines while training accuracy continues to rise, the model may be overfitting.

5.1.2 Loss

Loss quantifies how far the model's predictions are from the true target values. It is computed using a *loss function* (or *cost function*) that assigns a penalty to incorrect or inaccurate predictions. The goal of training is to minimize this loss function through iterative optimization, typically via gradient descent or one of its variants.

For example, in regression tasks, the most common loss function is the *Mean Squared Error (MSE)*, defined as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2,$$

where y_i represents the true value and $\hat{y}_i$ the predicted output. In classification tasks, loss functions such as *Cross-Entropy Loss* or *Negative Log-Likelihood* are widely used to measure the divergence between the predicted probability distribution and the true labels.

The *loss curve* typically decreases as training progresses, indicating that the model is minimizing its prediction errors. However, a widening gap between training and validation loss often signals overfitting, whereas consistently high loss on both sets may indicate underfitting or inappropriate model complexity.

In summary, accuracy and loss are fundamental diagnostic tools in machine learning. Their joint analysis provides valuable feedback on model convergence, learning efficiency, and generalization performance, guiding practitioners in fine-tuning architectures and optimization strategies.

5.2 Precision & Recall

While accuracy provides a general indication of model performance, it can be misleading in situations where data are imbalanced—when certain classes occur much more frequently than others. In such cases, metrics like *precision* and *recall* offer a more informative and nuanced evaluation of classification models, especially in binary or multi-class settings where the cost of false positives and false negatives differs.

Precision and recall are often analyzed together to assess how well a model identifies positive instances without introducing excessive misclassifications. Their trade-off is frequently visualized through a *precision–recall curve*, which illustrates the relationship between these two quantities as the decision threshold varies.

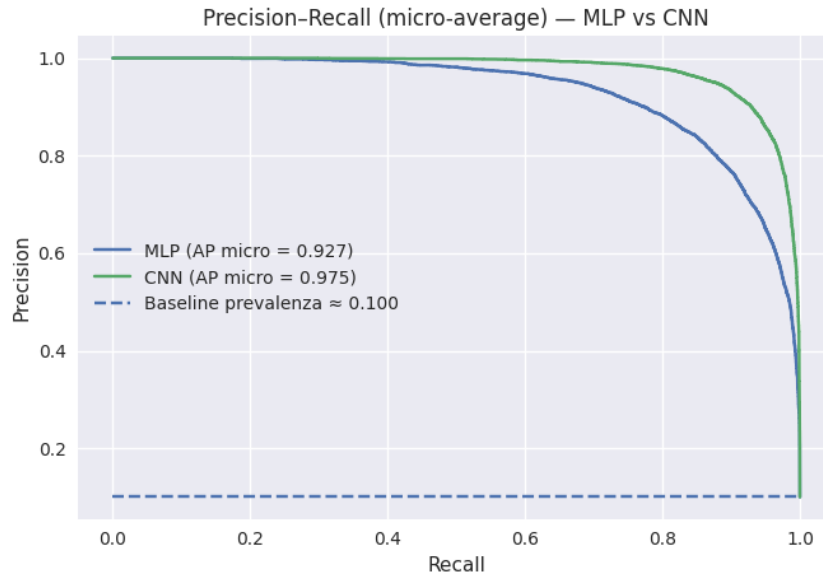


Figure 5.2: Precision Recall Curve

5.2.1 Precision

Precision measures the proportion of positive predictions that are actually correct. It answers the question: *Of all instances predicted as positive, how many are truly positive?* Mathematically, precision is defined as:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}.$$

High precision indicates that the model makes few false positive errors—it is “cautious” in labeling samples as positive. However, a model with very high precision may miss many actual positives if it becomes too conservative, leading to low recall.

In the precision–recall curve, precision typically decreases as recall increases. This occurs because, as the model becomes more inclusive (labeling more instances as positive), it also tends to introduce more false positives.

5.2.2 Recall

Recall, also known as *sensitivity* or *true positive rate*, measures the proportion of actual positives that the model successfully identifies. It answers the question: *Of all true positive instances, how many did the model correctly detect?* Formally, recall is defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}.$$

A high recall value indicates that the model captures most of the true positives but may include more false positives. In many real-world applications—such as medical diagnosis or fraud detection—high recall is critical, as missing a positive case can be costly or dangerous.

5.2.3 Confusion Matrix

The *confusion matrix* is a fundamental tool for evaluating the performance of classification models. It provides a detailed breakdown of the model's predictions compared to the actual ground truth labels, allowing practitioners to understand not only how many predictions were correct, but also the types of errors the model made.

In a binary classification setting, the confusion matrix is a 2×2 table that summarizes the counts of true and false predictions for both classes:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Here:

- **True Positives (TP):** Instances correctly classified as belonging to the positive class.
- **True Negatives (TN):** Instances correctly classified as belonging to the negative class.
- **False Positives (FP):** Instances incorrectly classified as positive (Type I error).
- **False Negatives (FN):** Instances incorrectly classified as negative (Type II error).

From the confusion matrix, a variety of performance metrics can be derived, including:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

These relationships make the confusion matrix an essential foundation for most evaluation metrics in classification problems.

For *multi-class classification*, the confusion matrix extends to an $N \times N$ table, where N is the number of classes. Each row represents the actual class, and each column represents the predicted class. Diagonal entries correspond to correct predictions, while off-diagonal entries indicate misclassifications between specific classes. Visualizing the confusion matrix as a heatmap can help reveal systematic biases—for instance, when the model consistently confuses certain categories due to similarity in features or insufficient training examples.

Confusion Matrix (counts)

T-shirt/top	862	1	14	35	2	1	79	0	6	0
Trouser	1	973	4	16	4	0	2	0	0	0
Pullover	20	0	865	10	65	0	40	0	0	0
Dress	12	6	9	928	21	0	24	0	0	0
Coat	0	0	26	24	914	0	36	0	0	0
Sandal	0	0	0	0	0	978	0	18	0	4
Shirt	114	3	46	33	87	0	711	0	6	0
Sneaker	0	0	0	0	0	9	0	972	1	18
Bag	0	1	2	3	1	2	3	0	988	0
Ankle boot	0	0	0	1	0	5	1	32	1	960

Predicted

Figure 5.3: Confusion Matrix for fashion MNIST dataset

In practice, confusion matrices are often used alongside precision–recall and ROC curves to provide a comprehensive understanding of a model’s behavior. They allow practitioners to identify which types of errors are most common, assess the balance of performance across classes, and guide future steps in data collection, model design, or hyperparameter tuning.

In summary, the confusion matrix is not merely a diagnostic tool but a window into the decision-making process of a classifier. It enables a granular analysis of model performance, transforming raw prediction counts into actionable insights for model refinement and validation.

5.2.4 Precision–Recall Trade-off and Curve Interpretation

Precision and recall are inherently linked: improving one often comes at the expense of the other. By adjusting the model’s decision threshold, one can balance the two metrics depending on the specific requirements of the application. The *precision–recall curve* plots recall on the x-axis and precision on the y-axis, summarizing performance across all possible thresholds.

A model that perfectly separates classes would achieve a precision of 1.0 across all recall levels, resulting in a flat line at the top of the plot. In contrast, a random classifier produces a curve close to the baseline, indicating that precision falls rapidly as recall increases. The area under the precision–recall curve (AUC–PR) provides a scalar summary of overall performance, especially useful when comparing different models on imbalanced datasets.

In summary, precision and recall offer complementary perspectives on model performance. Precision emphasizes correctness among positive predictions, while recall emphasizes completeness in capturing all positive instances. Together, they form the foundation for the *F1-score*, a harmonic mean that provides a single, balanced measure of both precision and recall—allowing for a more holistic evaluation of classification quality.

5.3 F1-Score

The *F1-score* is a widely used metric that combines *precision* and *recall* into a single measure of a model’s effectiveness. It is particularly valuable when dealing with imbalanced datasets, where accuracy alone can be misleading, and when both false positives and false negatives carry significant consequences.

Precision measures how many of the predicted positive instances are truly positive, while recall measures how many of the actual positive instances were correctly identified. Since these two metrics often trade off against each other, the F1-score provides a balanced summary by computing their harmonic mean:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

This formulation ensures that a high F1-score can only be achieved when both precision and recall are reasonably high. If either precision or recall is low, the F1-score will also decrease sharply, making it more sensitive to performance imbalances than a simple arithmetic mean.

The F1-score ranges between 0 and 1, where 1 indicates perfect precision and recall, and 0 indicates the worst possible performance. It is especially useful when the cost of false positives and false negatives is roughly equal, or when the dataset exhibits skewed class distributions.

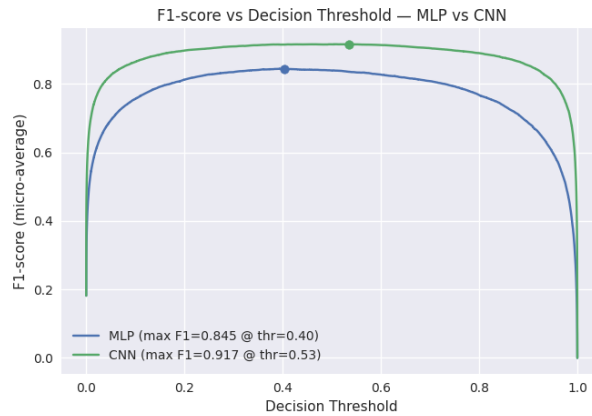


Figure 5.4: F1 score graph with threshold

In a binary classification context, a single F1-score value summarizes model performance. However, for multi-class problems, it can be generalized using different averaging strategies:

- **Macro F1-score:** Computes the F1-score independently for each class and then takes the unweighted mean. It treats all classes equally, regardless of their frequency.
- **Micro F1-score:** Aggregates all true positives, false positives, and false negatives across classes before computing the F1-score. It gives more weight to frequent classes.
- **Weighted F1-score:** Computes the class-specific F1-scores and averages them proportionally to the number of true instances in each class. This approach balances performance evaluation according to class importance.

The *F1-score curve* can also be plotted against varying classification thresholds to visualize the trade-off between precision and recall. By adjusting the decision threshold, practitioners can tune the balance between sensitivity (recall) and specificity (precision) to best fit the requirements of a specific application—such as maximizing recall in medical diagnosis or maximizing precision in spam detection.

In summary, the F1-score provides a robust and interpretable single-number metric for evaluating classification performance, especially in the presence of imbalanced data. It captures the delicate equilibrium between precision and recall, offering a more truthful representation of model effectiveness than accuracy alone. For this reason, it is a standard metric in research benchmarks and real-world machine learning evaluations alike.

Chapter 6

Implementation & Code Patterns

Chapter 7

Advanced Topics